

Introduction à QGIS pour les humanités numériques

1. Introduction générale aux SIG et à QGIS.
2. L'interface de QGIS.
3. Création, (re)utilisation et Visualisation des données vectorielles et trames.

Pause

4. Géoréférencement
5. Spatallite
6. Analyses spatials et création des cartes

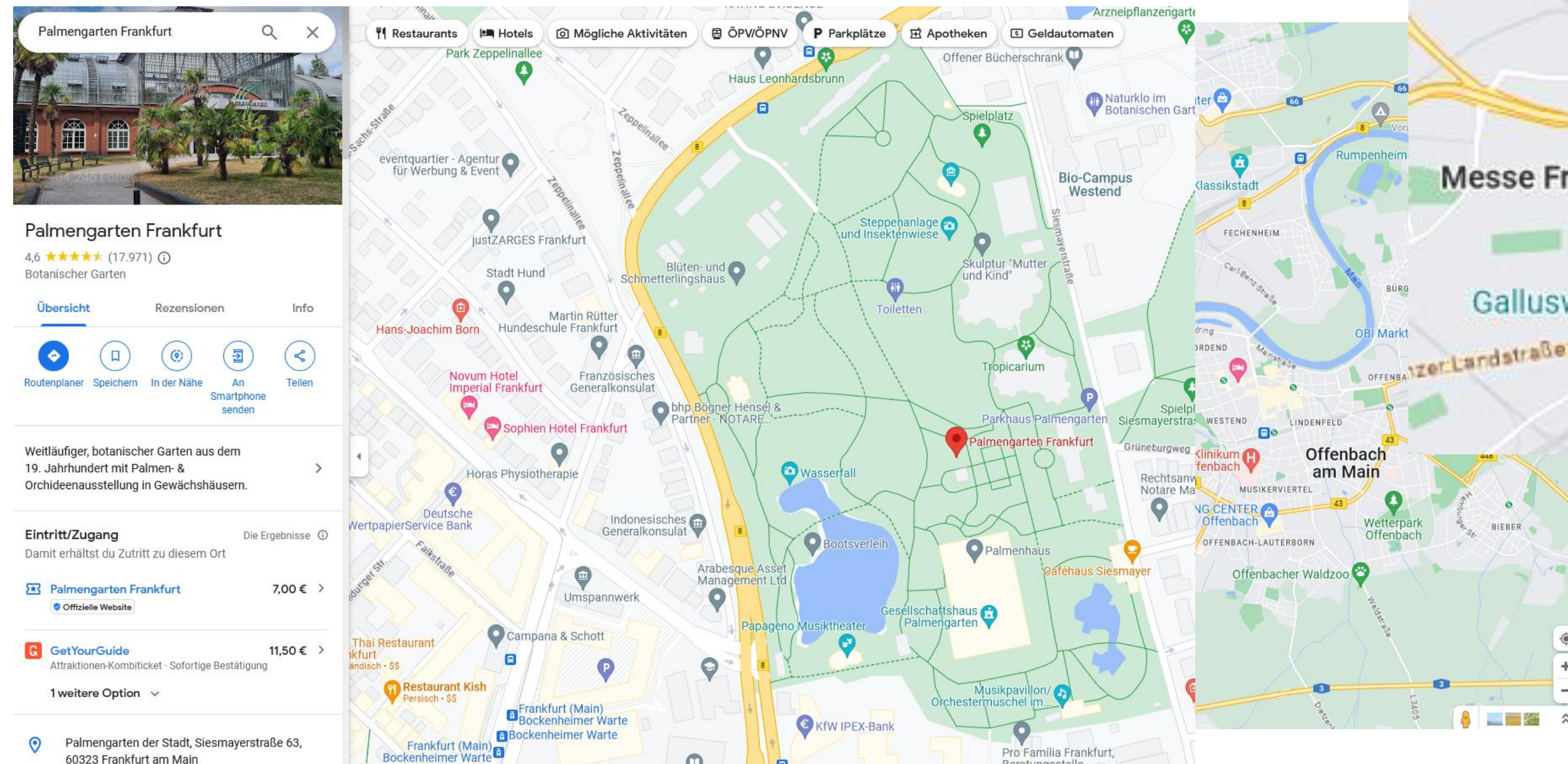
1. Introduction générale aux SIG et à QGIS

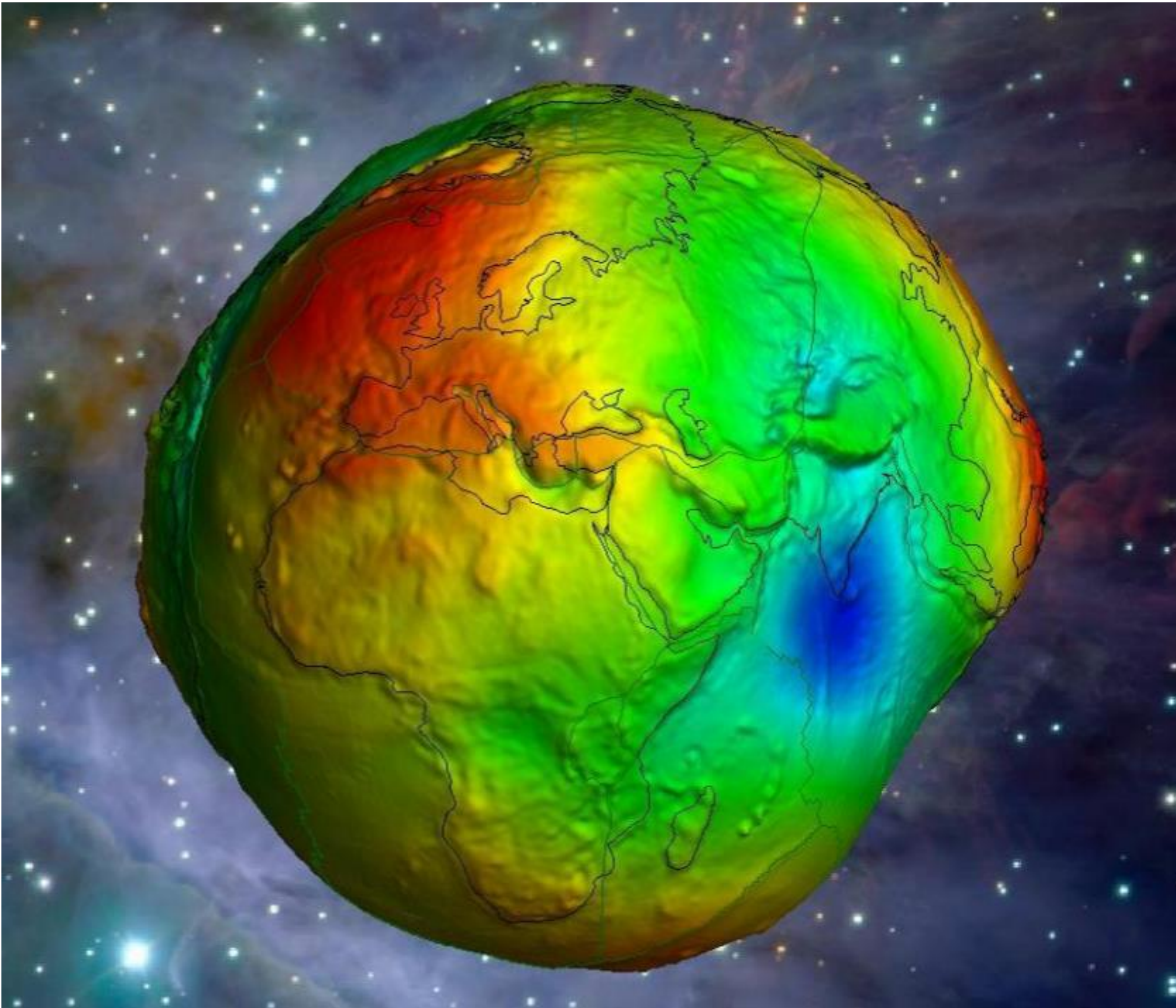
GIS – Geographical Information System

Base de données

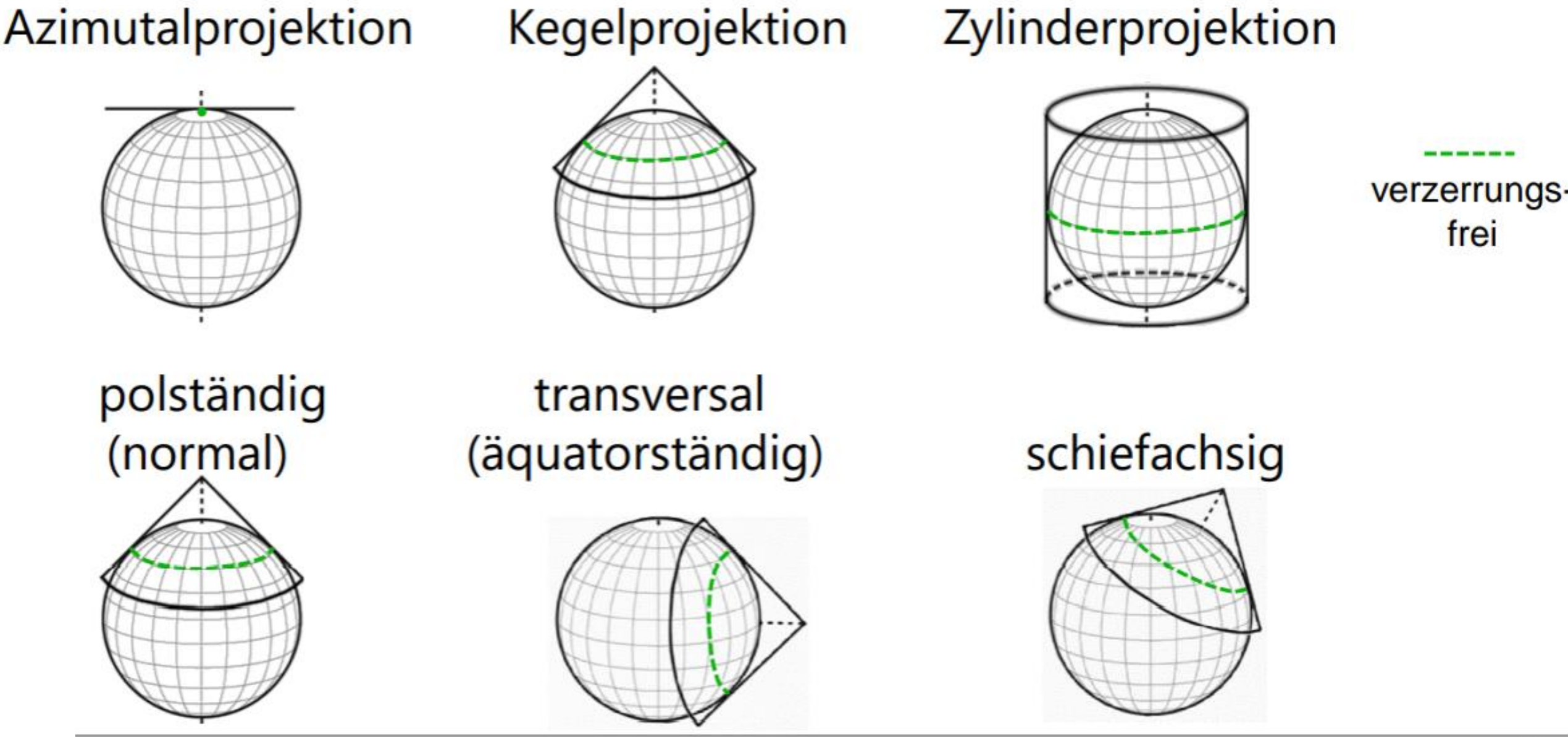
- Objects spatials (Points, Lignes, Aréals)
- Attributes des objects spatials (âge, dimensions, nom, région d'origine)

→ Le GIS peut connecter ces informations



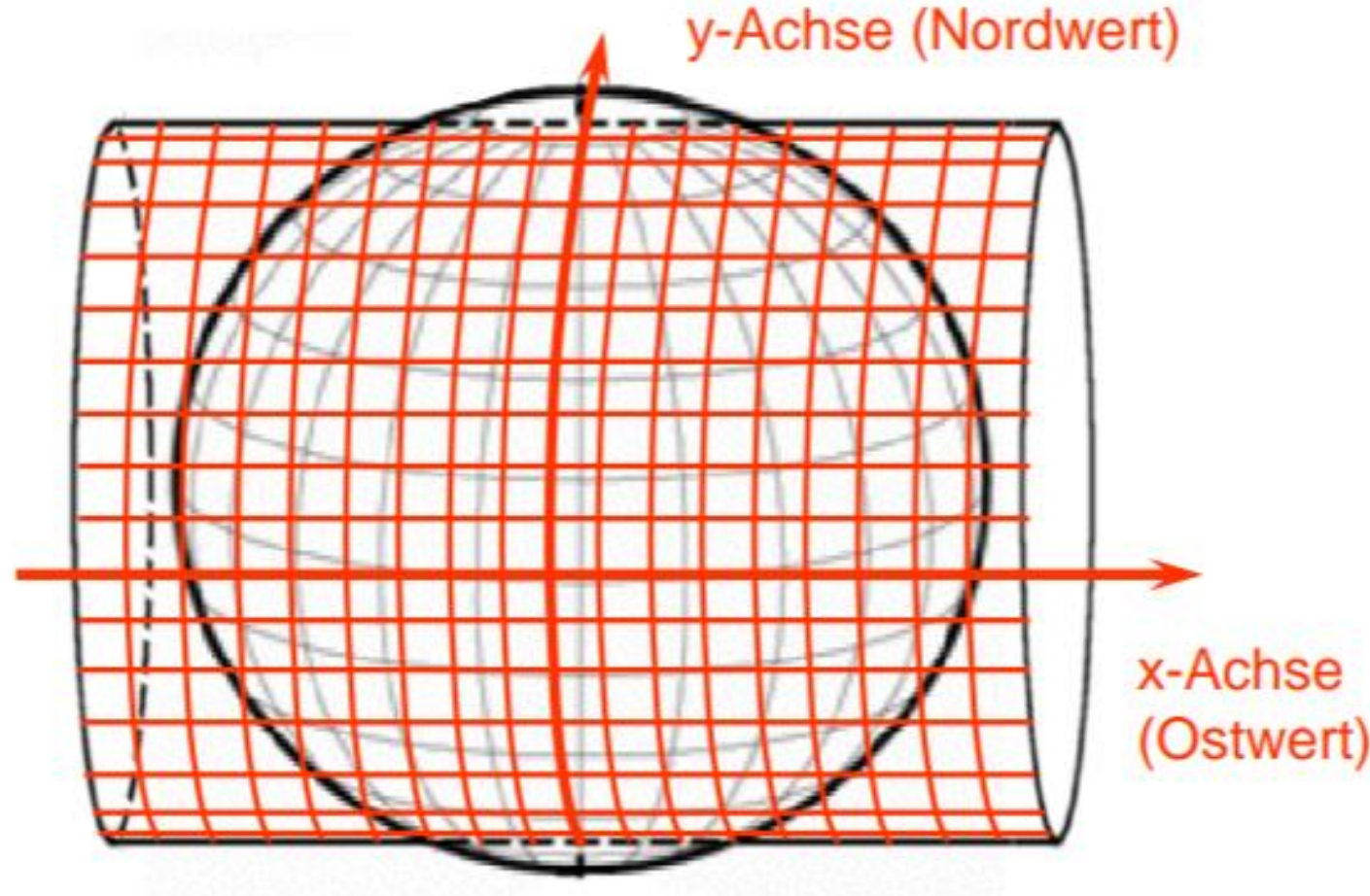


– 3 **Grundtypen** in 3 verschiedenen **Lagen**:



WGS84-Ellipsoid and Date
→ worldwide

GRS80-Ellipsoid with
ETRS89-Date (UTM)
→ europe



Utilised in Europe: ETRS 89 / UTM zone ???

- Separates the world into 60 stripes that are further differentiated
 - (Slight overlay to improve coverage)
 - The system is keeping the angles of the real world:
 - 90° angle on earth is a 90° angle in the projection
 - Distances will be slightly incorrect
 - Stripes in the northern and southern hemisphere (S/N)
- Germany and Switzerland: ETRS 89 / UTM zone 32N
 → Western part of Geneva is ETRS 89 / UTM zone 31N

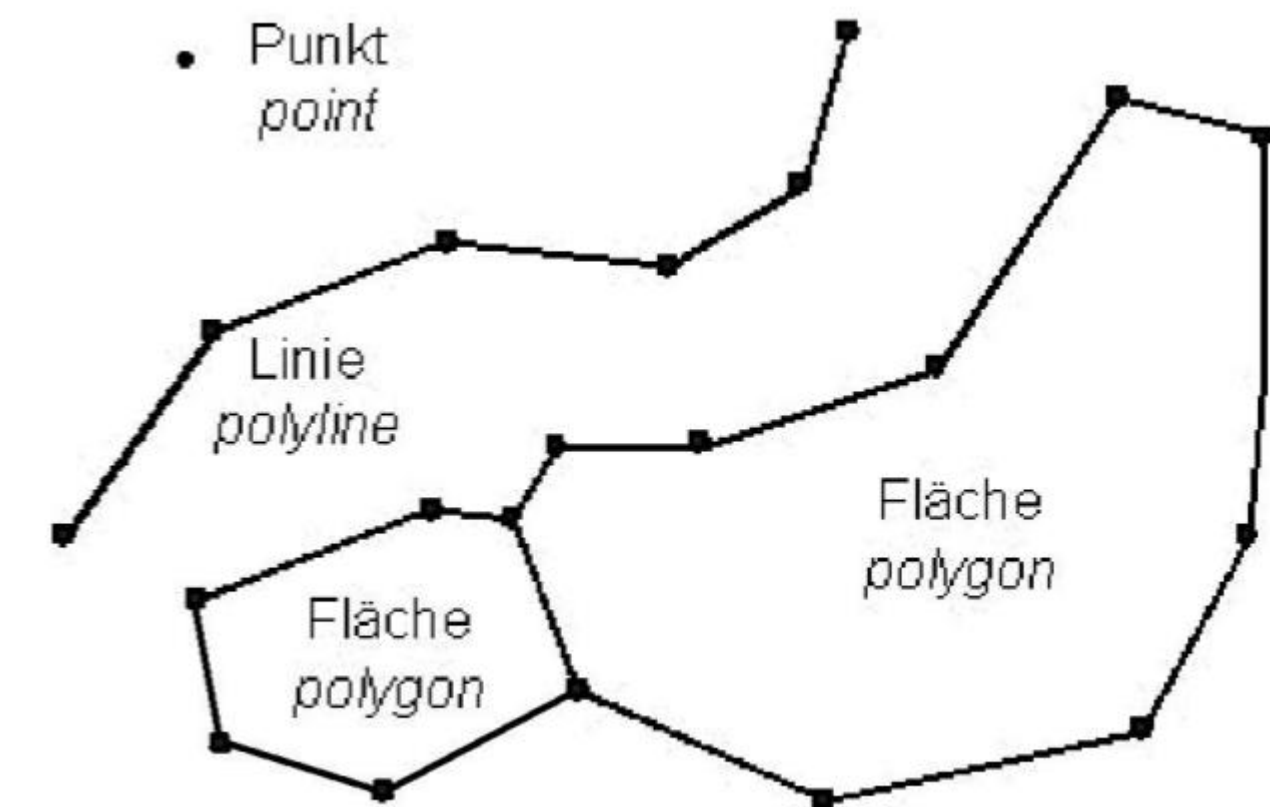


Gemeinfrei, <https://commons.wikimedia.org/w/index.php?curid=1128616>

Discrete objects

→ Points with coordinates

Informations are transmitted via attributes

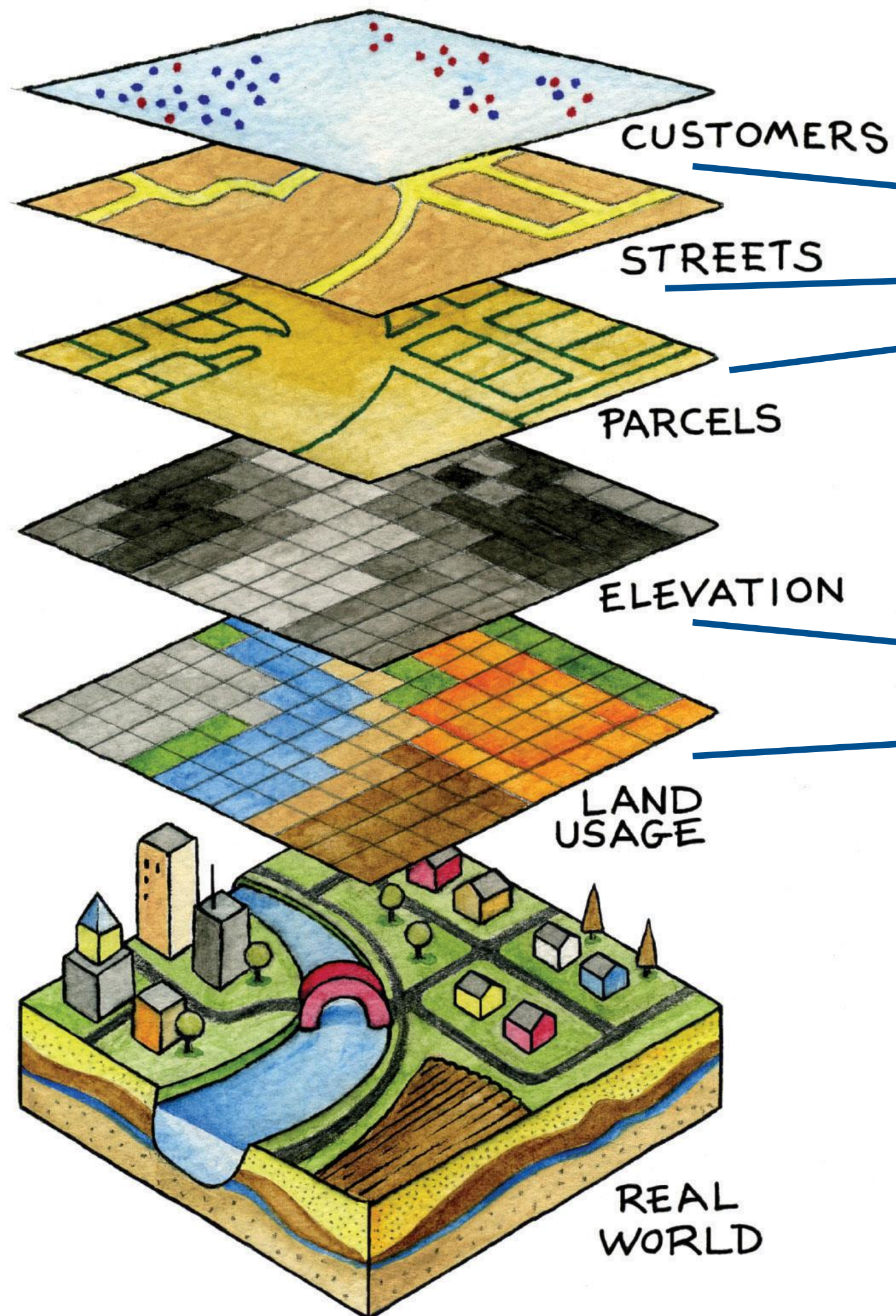


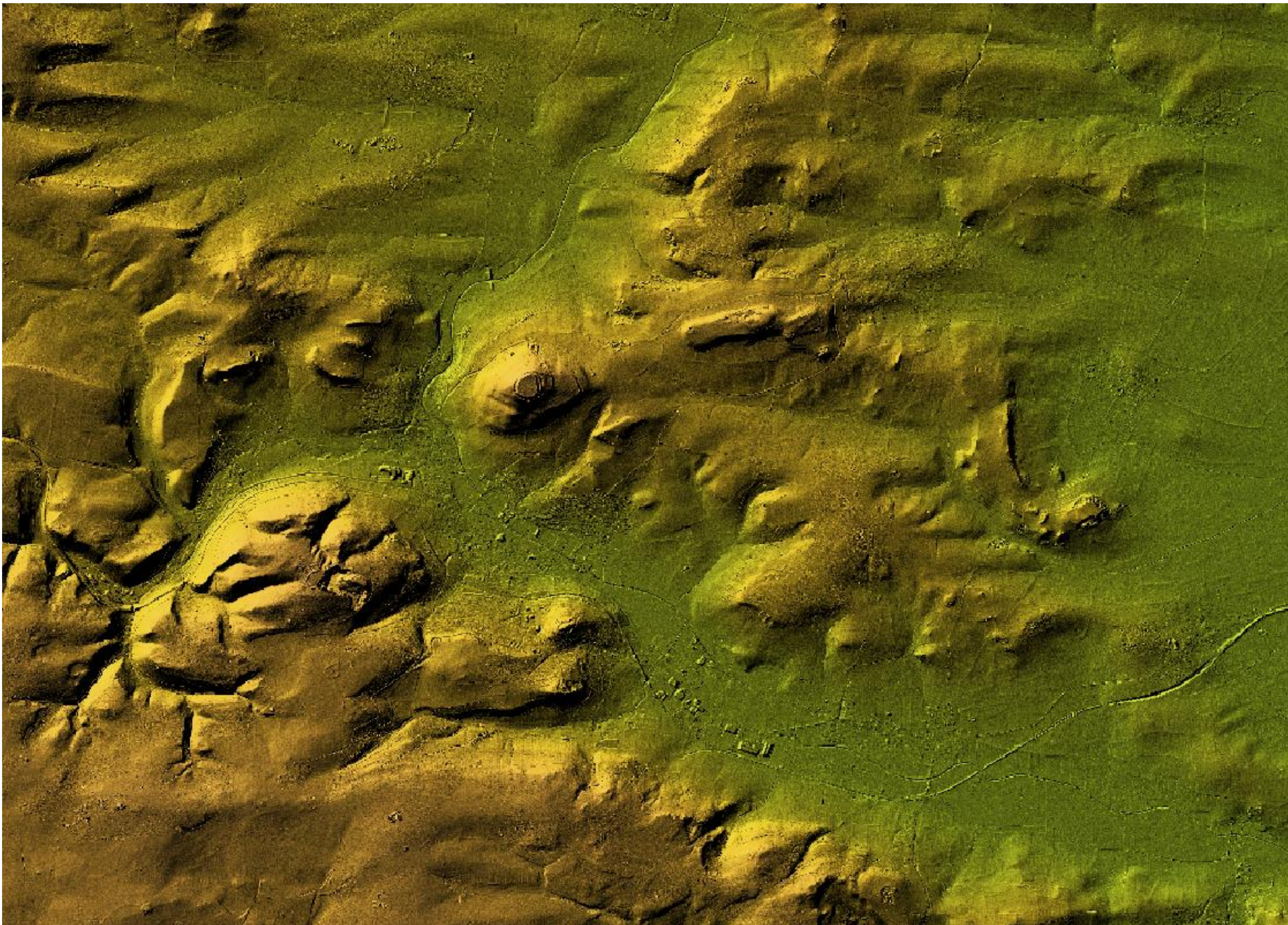
vector (.shp)

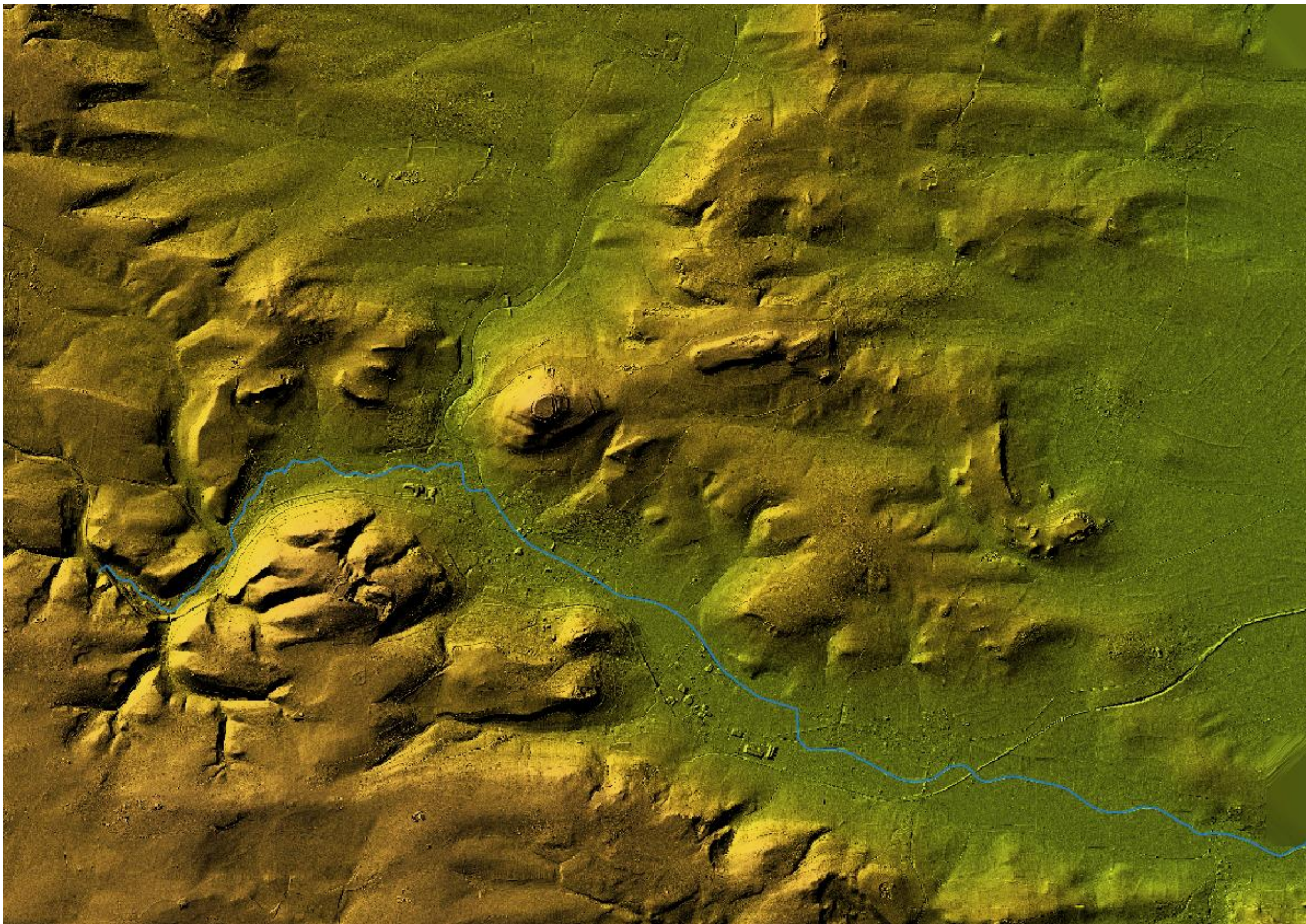
raster (.tif)

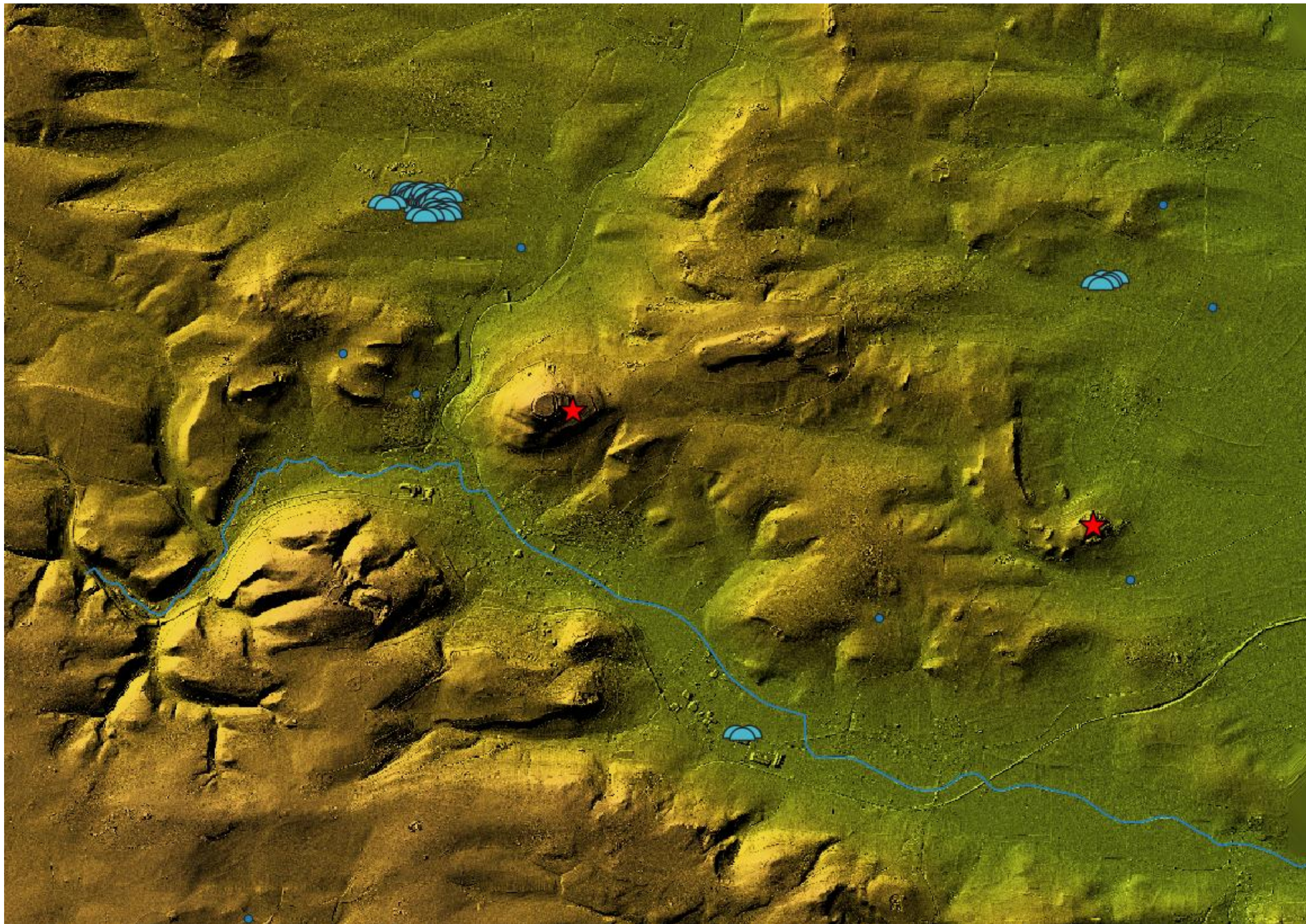
Continuous objects are displayed as areas by being separated into raster squares

Informations are transmitted via the color of the pixel

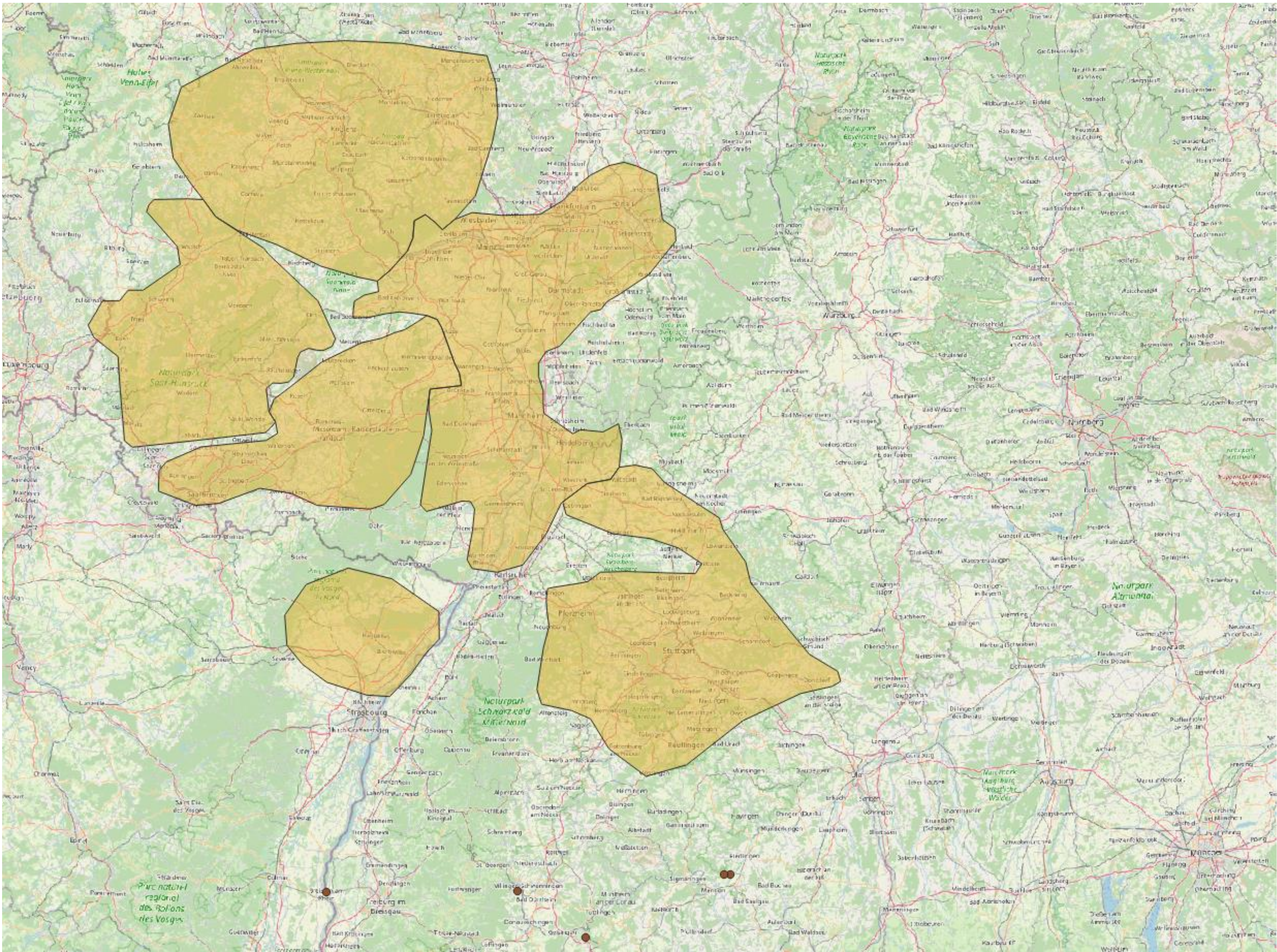








	ogc_fid	ident	archivkenn	anschrift	objekttyp	objektklas	zeitstell	eigzeit	listentext	ergebnis	shape_id	layer	path	geprueft	zeitstufe	Literatur
1	118	103030361	DEWA008	Aalen, Dewang...	Siedlung	Siedlung	Hallstattzeit	Hallstattzeit	NULL	2	ARC	Aalen_M	C:/Users/Jonat...	Nein	NULL	NULL
2	282	96961358	EBNA001	Aalen-Niesitz, E...	Grabhuegelfeld	Grabhuegel	Hallstattzeit	Hallstattzeit	NULL	1	ARC	Aalen_M	C:/Users/Jonat...	Nein	NULL	NULL
3	436	95268651	WALD005	Aalen-Waldhau...	Grabhuegelfeld	Grabhuegel	Hallstattzeit	Hallstattzeit	NULL	4	ARC	Aalen_M	C:/Users/Jonat...	Nein	NULL	NULL
4	437	96993526	WALD004	Aalen-Waldhau...	Grabhuegelfeld	Grabhuegel	Hallstattzeit	Hallstattzeit	NULL	5	ARC	Aalen_M	C:/Users/Jonat...	Nein	NULL	NULL
5	440	108157580	WALD017	Aalen, Waldhau...	Grabhuegelfeld	Grabhuegel	Hallstattzeit	Hallstattzeit	NULL	17	ARC	Aalen_M	C:/Users/Jonat...	Nein	NULL	NULL
6	452	96993504	WALD003	Aalen-Waldhau...	Grabhuegelfeld	Grabhuegel	Hallstattzeit	Hallstattzeit	NULL	3	ARC	Aalen_M	C:/Users/Jonat...	Nein	NULL	NULL
7	476	99545078	WASS010	Aalen-Wasseral...	Grabhuegelfeld	Grabhuegel	Hallstattzeit	Hallstattzeit	NULL	NULL	ARC	Aalen_M	C:/Users/Jonat...	Nein	NULL	NULL
8	632	96988156	BOPF001, L7128...	Bopfingen, Bop...	Hoehensiedlung	Siedlung	Urnenfelderzeit,...	Neolithikum, Br...	NULL	1	ARC	Bopfingen_M	C:/Users/Jonat...	Ja	Bz A2, Bz B1, H...	Knoll 2021, Kat. ...
9	649	96961588	BOPF010	Bopfingen-Floc...	Kreisgraben	Erdwerk	Mittlere Hallsta...	Hallstattzeit	NULL	2	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
10	689	103017915	KERK009	Bopfingen, Ker...	Siedlungsreste	Siedlung	Spaete Hallstatt...	Hallstattzeit	NULL	5	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
11	698	96961380	KERK002	Bopfingen-Mei...	Grabhuegelfeld	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	1	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
12	700	96990685	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
13	701	96990686	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
14	702	96990687	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
15	703	96990688	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
16	704	96990695	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
17	705	96990696	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
18	706	96990697	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
19	707	96990698	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
20	708	96990683	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
21	710	96990684	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
22	711	96990701	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
23	712	96990702	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
24	713	96990703	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL
25	714	96990704	KERK002	Bopfingen-Mei...	Grabhuegel	Grabhuegel	Mittlere Hallsta...	Hallstattzeit	NULL	NULL	ARC	Bopfingen_M	C:/Users/Jonat...	Nein	NULL	NULL

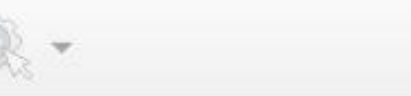


Fundort	Beginn Chr	Ende Chr
Heuneburg	Ha D1	Ha D3
Magdalenenberg	Ha C	Ha D2
Hohmichele	Ha C	Ha D2
Mauenheim	Ha C mittel	Ha D2
Münsterberg v...	NULL	NULL

2. L'Interface de QGIS

Untitled Project — QGIS

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

Browser

- ★ Favorites
- ▶ Spatial Bookmarks
- ▶ Home
- ▶ C:\ (Windows-SSD)
- GeoPackage
- ▶ Spatialite
 - Grabungsdokumentation.sqlite
- PostgreSQL
- SAP HANA
- STAC
- MS SQL Server
- Oracle
- WMS/WMTS

Recent Projects


Grabungsdokumentation Ohrenberg
C:/Users/jschm/OneDrive - Johann Wolfgang Goethe Universität/Ohrenberg/10_GIS/Ohrenberg-GIS/Ohrenberg-GIS/Ohrenberg-GIS-neu.qgz

News

New QGIS Map Gallery

QGIS has a new map gallery! We are looking for submissions of beautiful maps made using QGIS. The maps will be displayed on the main web site at <https://qgis.org> if approved. Our goal is to show off the power and versatility of QGIS to the world. If you have a great map to share, **double click this article** to visit the gallery and start sharing your submissions!

QGIS User Conference 2025 will take place in June in Sweden

Save the date! The next global QGIS User Conference will be in Norrköping, Sweden from 2-3 June, 2025. Our contributor meeting will be directly afterwards from 4 to 6 June, 2025.

QGIS User Conference and Contributor meetings are the best place to engage with the

Project Templates

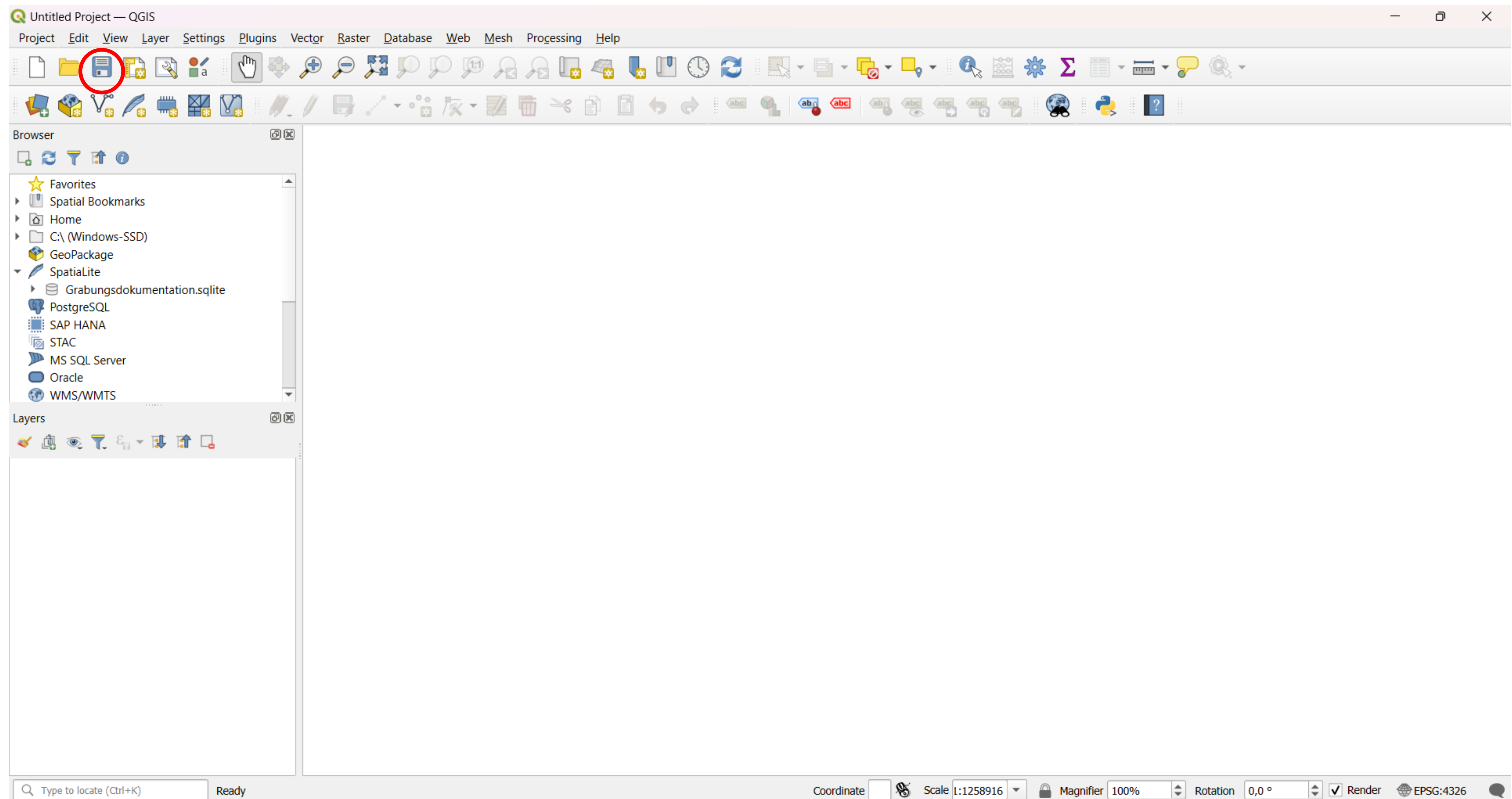
New Empty Project

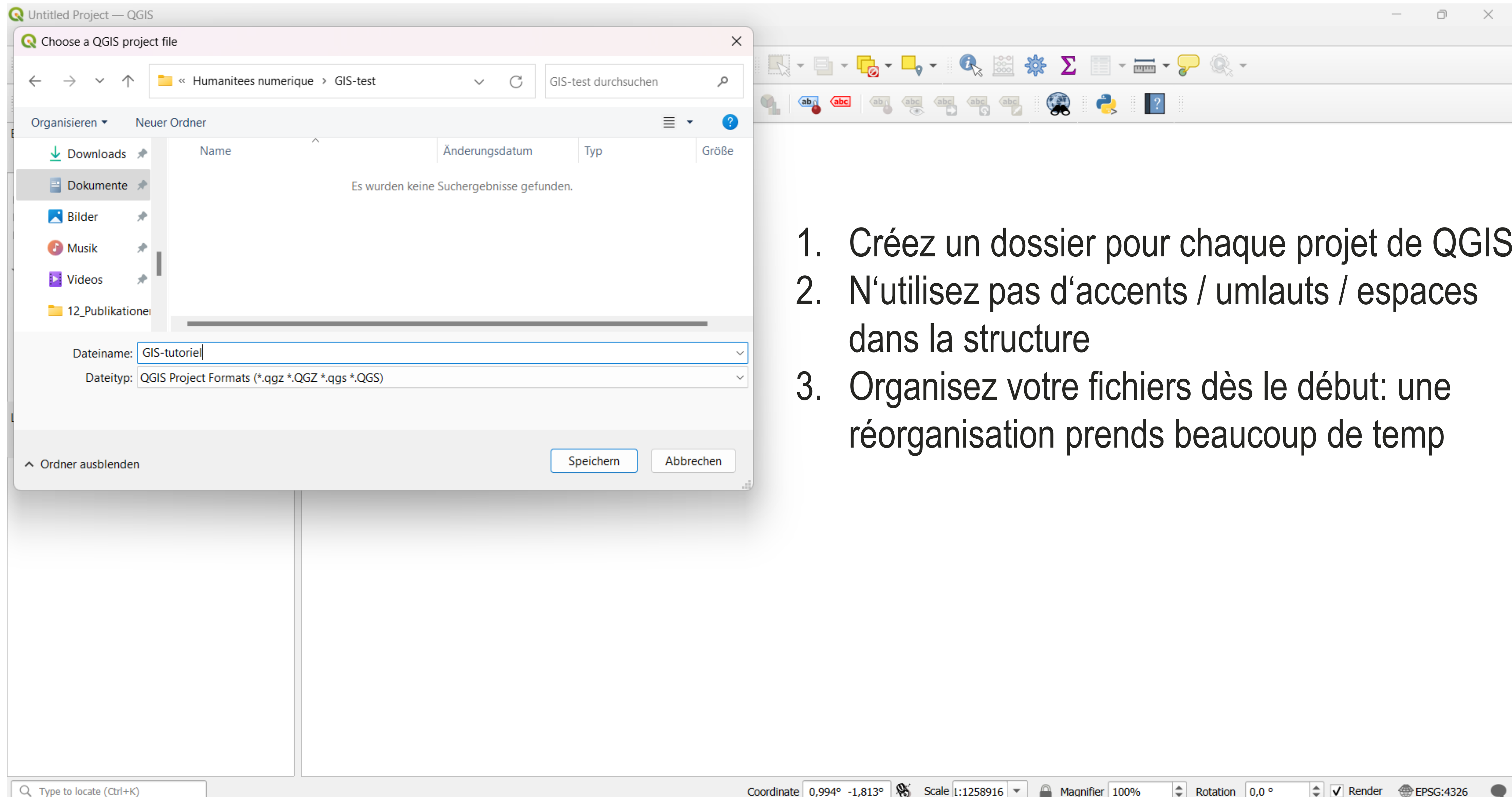
EPSG:4326 - WGS 84

New QGIS version available: Visit <https://download.qgis.org> to get your copy of version 3.40.5

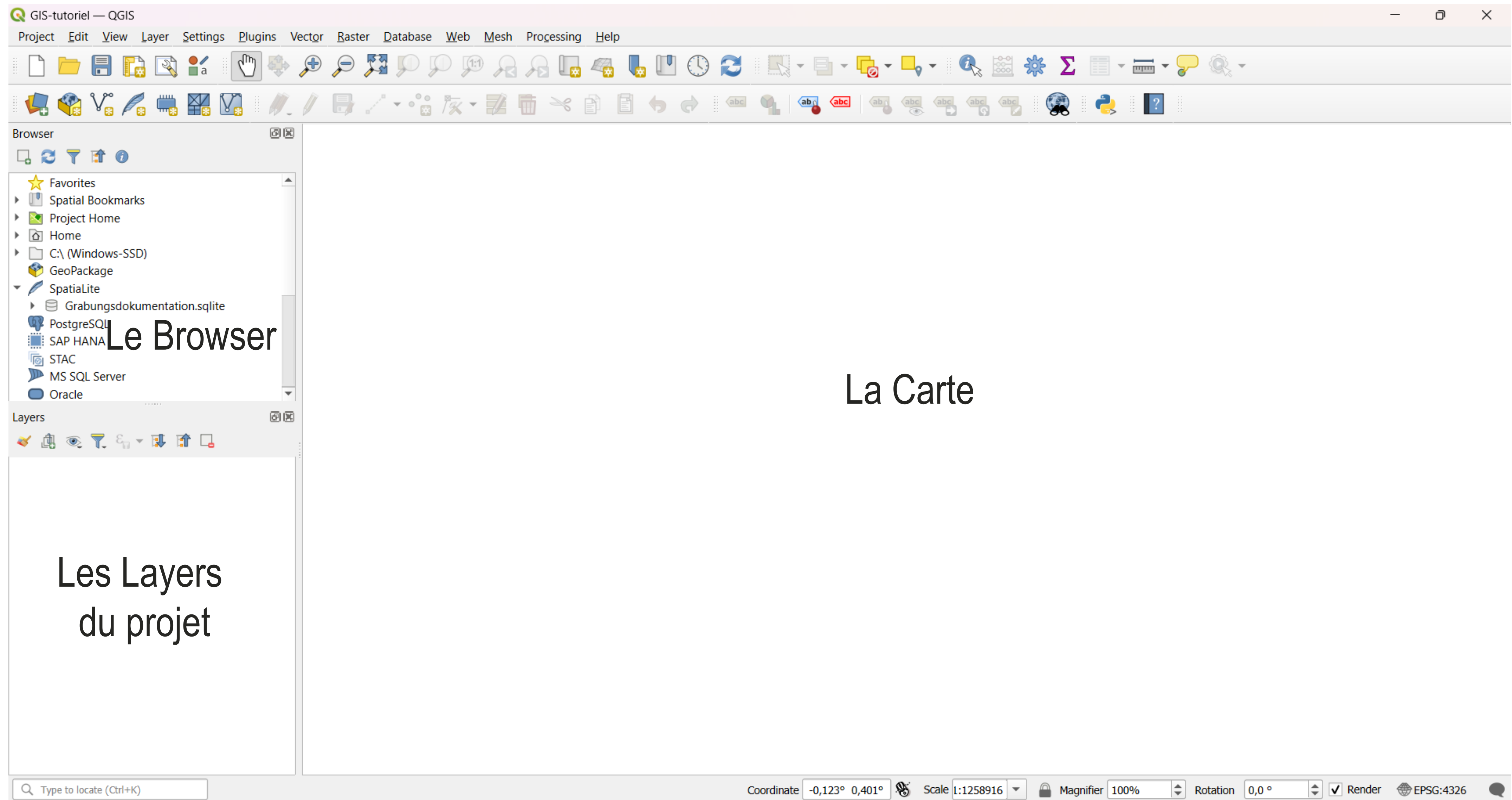
Type to locate (Ctrl+K) Ready

Coordinate  Scale :29854292  Magnifier 100%  Rotation 0,0 °  ☒ Render  EPSG:4326 





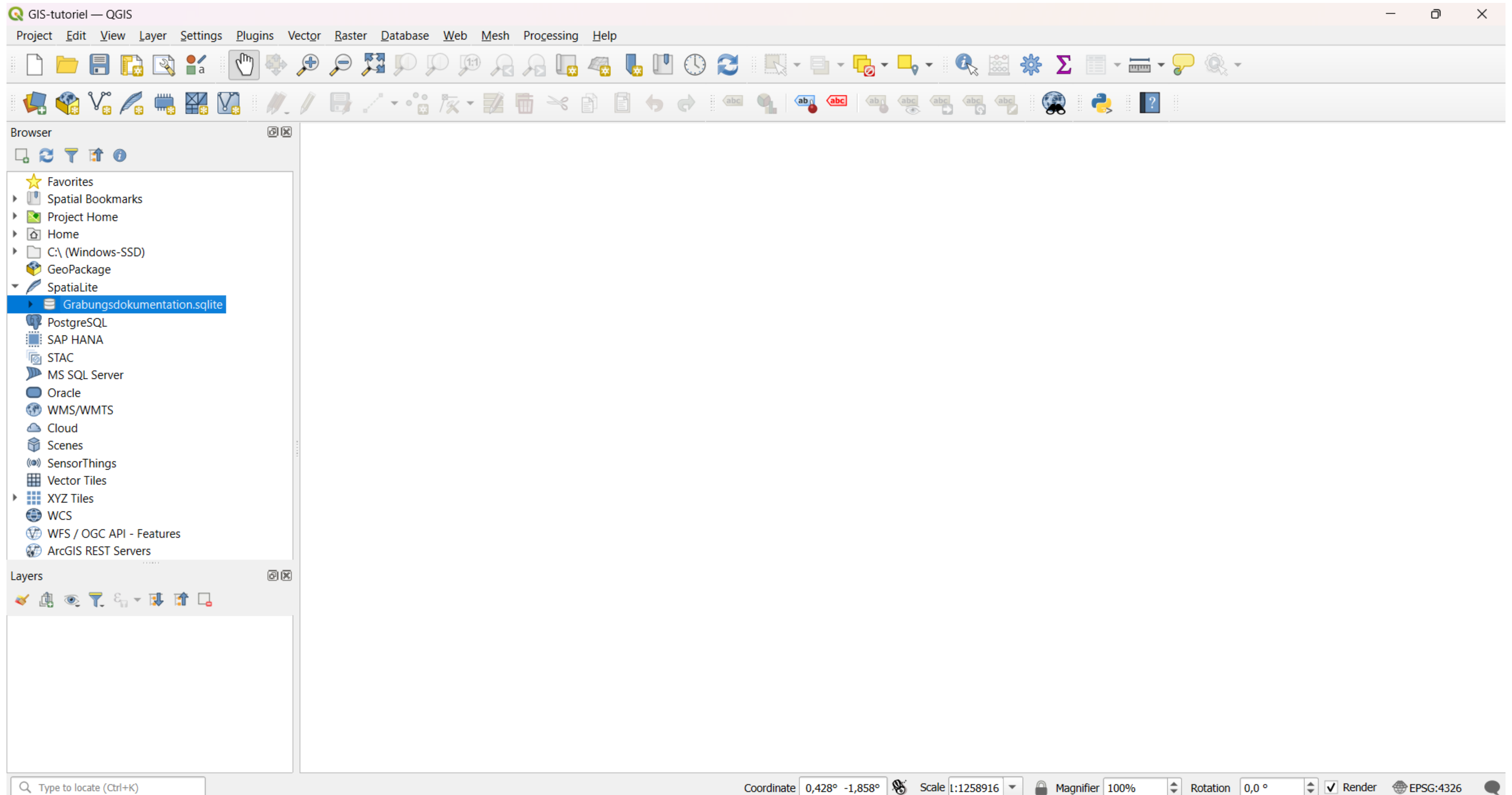
1. Créez un dossier pour chaque projet de QGIS
2. N'utilisez pas d'accents / umlauts / espaces dans la structure
3. Organisez votre fichiers dès le début: une réorganisation prends beaucoup de temp

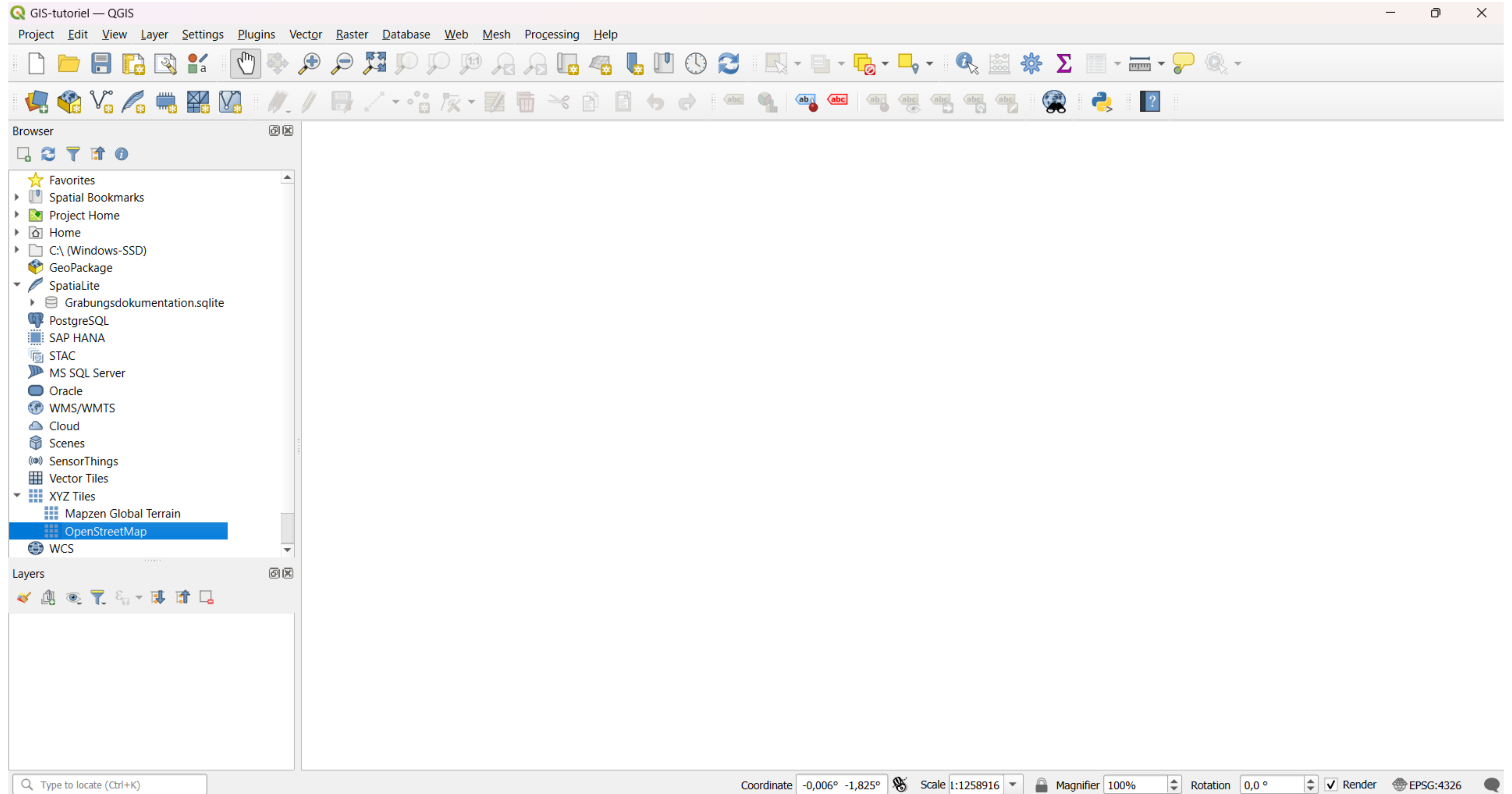


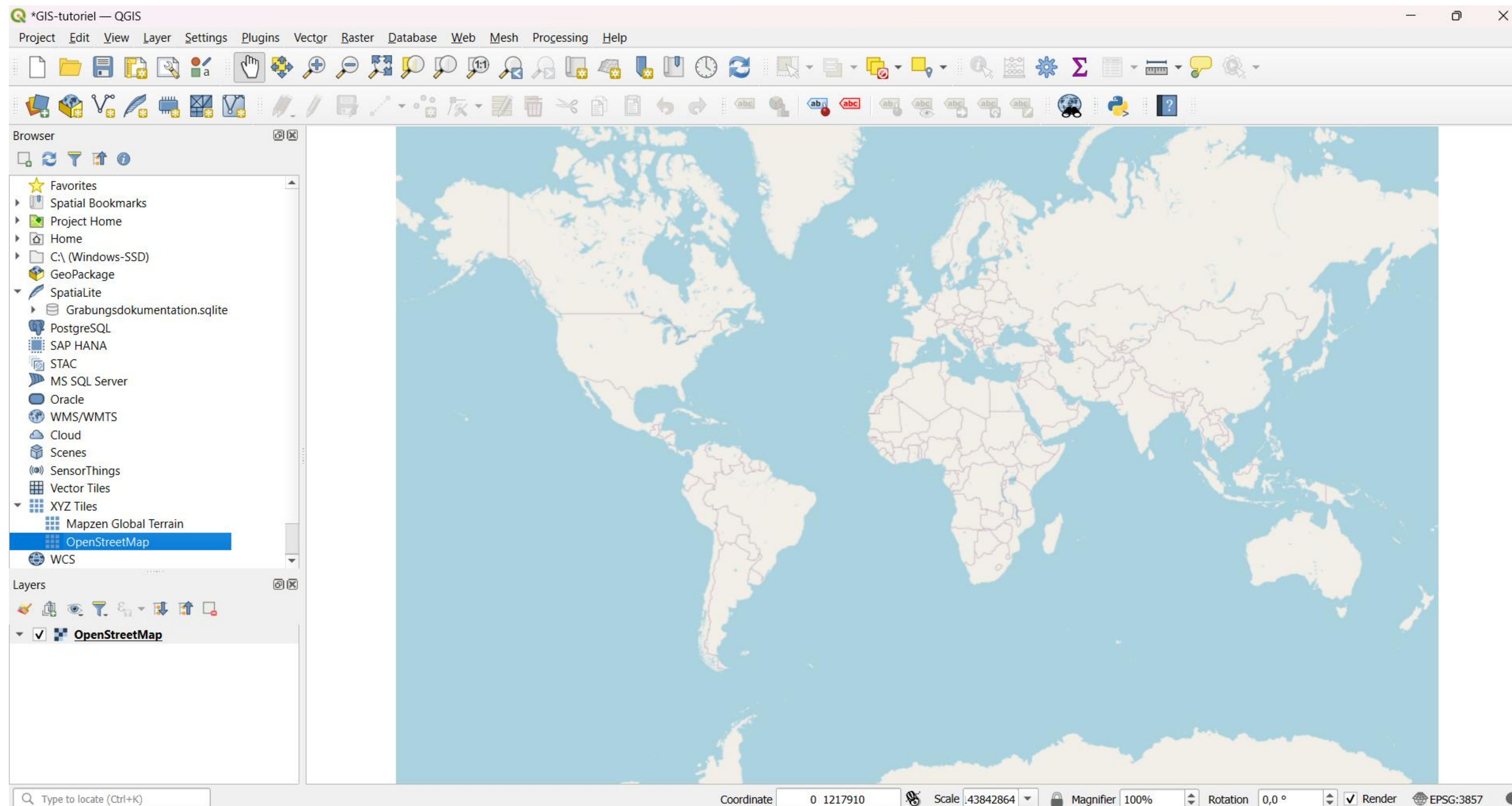
Les Bases
des données

Des données
utilisable

On peut
réorganiser
les panneaux







The screenshot shows the QGIS desktop environment. The main window is titled '*GIS-tutorial — QGIS'. The menu bar includes Project, Edit, View, Layer, Settings, Plugins, Vector, Raster, Database, Web, Mark, Processing, and Help. The Plugins menu is open, showing options: All, Installed, Not installed, Install from ZIP, and Settings. The 'All' option is selected, displaying a list of plugins. The 'OSM place search' plugin is highlighted. The details for this plugin are shown on the right side of the Plugins window. The details include the title 'OSM place search', a subtitle 'Location search helper', a description 'Based on OSM data, Nominatim online tool (https://operations.osmfoundation.org/policies/nominatim/)', a rating of 4 stars (146 rating vote(s), 474818 downloads), tags 'location search helper osm', more info links 'homepage bug tracker code repository', author 'Xavier Culos (Agence de l'eau Adour Garonne)', and available version '1.4.5 updated at 02.02.2023 16:09 Mitteleuropa'. The background map shows a portion of Europe.

*GIS-tutorial — QGIS

Project Edit View Layer Settings **Plugins** Vector Raster Database Web Mark Processing Help

Plugins | All (1497)

Search: osm pla

OSM place search

OSM place search

Location search helper

Based on OSM data, Nominatim online tool
(<https://operations.osmfoundation.org/policies/nominatim/>)

★★★★☆ 146 rating vote(s), 474818 downloads

Tags [location search helper](#) [osm](#)

More info [homepage](#) [bug tracker](#) [code repository](#)

Author Xavier Culos (Agence de l'eau Adour Garonne)

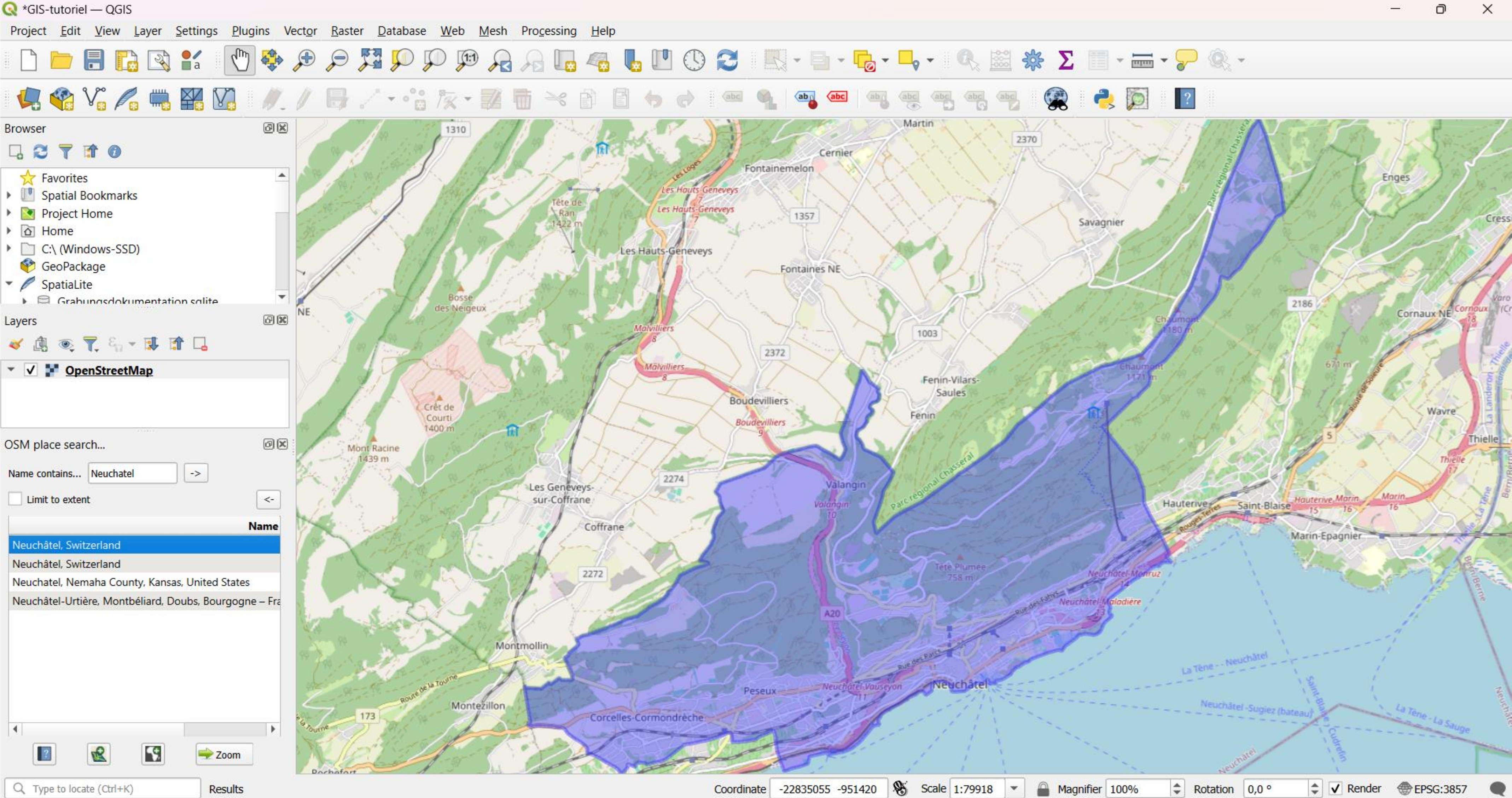
Available version (stable) [1.4.5](#) updated at 02.02.2023 16:09 Mitteleuropa

Upgrade All Install Plugin Close Help

Type to locate (Ctrl+K)

Coordinate: 11.557701 49.454004 Scale: 1:43642004 Magnifier: 100% Rotation: 0,0

Render EPSG:3857



QGIS-tutorial — QGIS

Project Edit View Layer Settings Plugins Vector

Browser

- ★ Favorites
 - ▶ Spatial Bookmarks
 - ▶ Project Home
 - ▶ Home
 - ▶ C:\ (Windows-SSD)
 - ▶ GeoPackage
 - ▶ SpatialLite
 - ▶ Grabungsdokumentation.sqlite

Layers

- ✓ OpenStreetMap

OSM place search...

Name contains... ->

☐ Limit to extent

Name
Neuchâtel, Switzerland
Neuchâtel, Switzerland
Neuchatel, Nemaha County, Kansas, United States
Neuchâtel-Urtière, Montbéliard, Doubs, Bourgogne – France

Project Properties — CRS

Project Coordinate Reference System (CRS)

☐ No CRS (or unknown/non-Earth projection)

Filter

Recently Used Coordinate Reference Systems

Coordinate Reference System	Authority ID
EPSG:4326 - WGS 84	EPSG:4326

Predefined Coordinate Reference Systems ☐ Hide deprecated CRSs

Coordinate Reference System	Authority ID
WGS 84 / Pseudo-Mercator	EPSG:3857
WGS 84 / World Mercator	EPSG:3395
WGS_1984_Web_Mercator	ESRI:102113
WGS_1984_Web_Mercator_Auxiliary_Sphere	ESRI:102100
WGS84 WEB MERCATOR SPHERIQUE (VISUALISATION)	IGNF:WGS84WMSV

WGS 84 / Pseudo-Mercator

Properties

- Units: meters
- Dynamic (relies on a datum which is not plate-fixed)
- Celestial body: Earth
- Based on *World Geodetic System 1984 ensemble* (EPSG:6326), which has a limited accuracy of **at best 2 meters**.
- Method: Mercator

WKT

```
PROJCRS["WGS 84 / Pseudo-Mercator",  
  BASEGEOGCRS["WGS 84",  
    ENSEMBLE["World Geodetic System 1984 ensemble",  
      MEMBER["World Geodetic System 1984 (Transit)"],  
      MEMBER["World Geodetic System 1984 (G730)"],  
      MEMBER["World Geodetic System 1984 (G873)"],  
      MEMBER["World Geodetic System 1984 (G1150)"],  
      MEMBER["World Geodetic System 1984 (G1674)"],  
      MEMBER["World Geodetic System 1984 (G1762)"],  
      MEMBER["World Geodetic System 1984 (G2139)"],
```

Rotation 0,0 ° ☒ Render EPSG:3857

Project Properties — CRS

Project Coordinate Reference System (CRS)

☐ No CRS (or unknown/non-Earth projection)

Filter:

Recently Used Coordinate Reference Systems

Coordinate Reference System	Authority ID
ETRS89 / UTM zone 32N	EPSG:25832

Predefined Coordinate Reference Systems ☐ Hide deprecated CRSs

Coordinate Reference System	Authority ID
ETRS89 / UTM zone 32N	EPSG:25832
ETRS89 / UTM zone 32N (N-E)	EPSG:3044
ETRS89/DREF91/2016 / UTM zone 32N	EPSG:10732

ETRS89 / UTM zone 32N

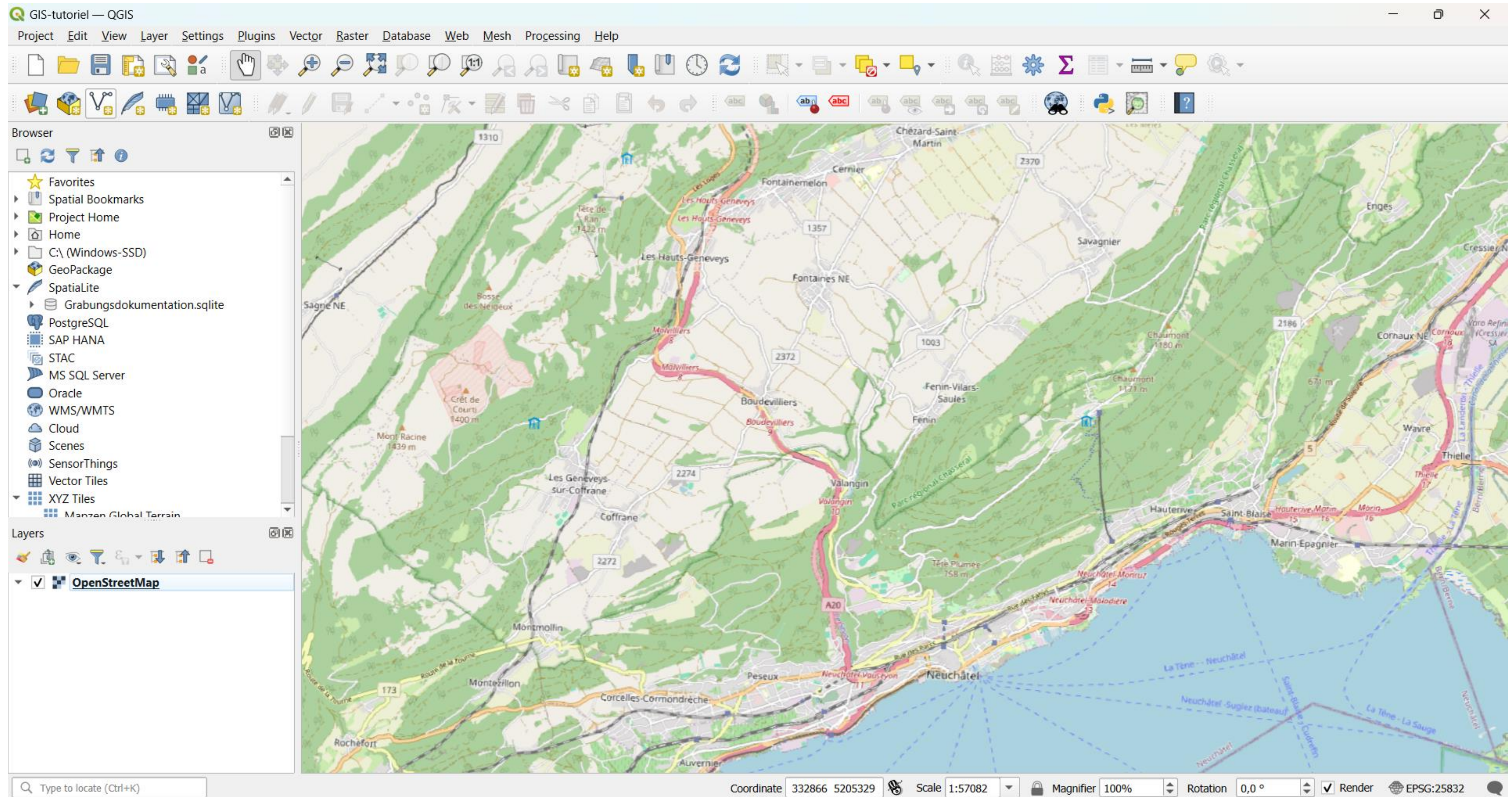
Properties

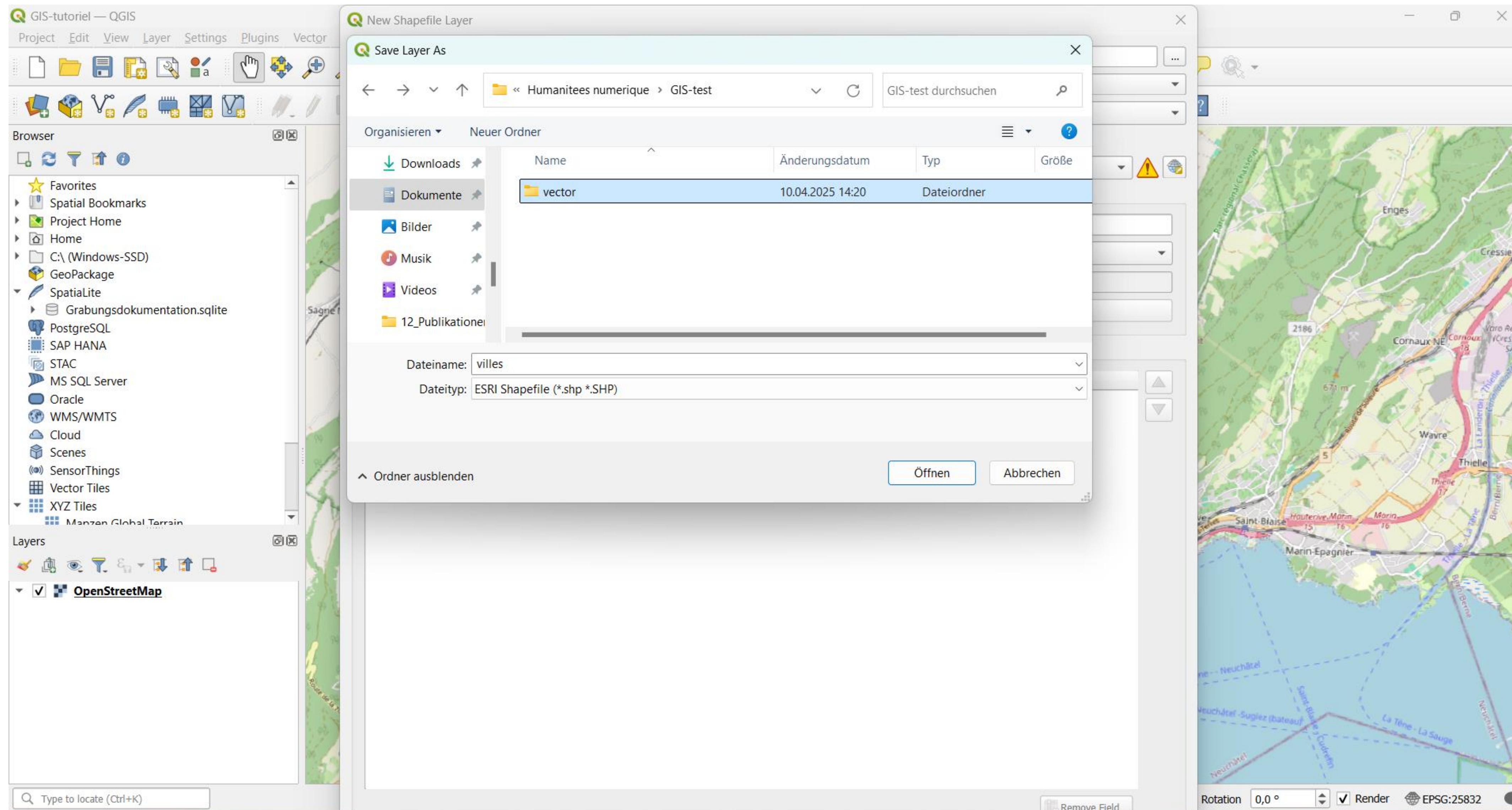
- Units: meters
- Static (relies on a datum which is plate-fixed)
- Celestial body: Earth
- Based on *European Terrestrial Reference System 1989 ensemble* (EPSG:6258), which has a limited accuracy of **at best 0.1 meters**.
- Method: Universal Transverse Mercator (UTM)

WKT

```
PROJCRS["ETRS89 / UTM zone 32N",
  BASEGEOGCRS["ETRS89",
    ENSEMBLE["European Terrestrial Reference System 1989 ensemble",
      MEMBER["European Terrestrial Reference Frame 1989"],
      MEMBER["European Terrestrial Reference Frame 1990"],
      MEMBER["European Terrestrial Reference Frame 1991"],
      MEMBER["European Terrestrial Reference Frame 1992"],
      MEMBER["European Terrestrial Reference Frame 1993"],
      MEMBER["European Terrestrial Reference Frame 1994"],
      MEMBER["European Terrestrial Reference Frame 1996"],
      MEMBER["European Terrestrial Reference Frame 1997"]
    ]
  ],
  CS[Cartesian],
  AXIS[Easting, EAST, METERS],
  AXIS[Northing, NORTH, METERS]
]
```


3. Création, (re)utilisation et Visualisation des données vectorielles et trames





GIS-tutorial — QGIS

Project Edit View Layer Settings Plugins Vector

Browser

- ★ Favorites
- ▶ Spatial Bookmarks
- ▶ Project Home
- ▶ Home
- ▶ C:\ (Windows-SSD)
- GeoPackage
- ▶ SpatialLite
- ▶ Grabungsdokumentation.sqlite
- PostgreSQL
- SAP HANA
- STAC
- MS SQL Server
- Oracle
- WMS/WMTS
- Cloud
- Scenes
- SensorThings
- Vector Tiles
- ▶ XYZ Tiles
- Manzen Global Terrain

Layers

- ✓ **OpenStreetMap**

Search: Type to locate (Ctrl+K)

New Shapefile Layer

File name: C:\Users\jschm\Documents\Humanitees numerique\GIS-test\vector\villes.shp

File encoding: System

Geometry type: Polygon

Additional dimensions:

New Field

Name:

Type: abc Text (string)

Length: 80 Precision:

Add to Fields List

Fields List

Name	Type	Length	Precision
id	Integer	10	

Remove Field

Rotation: 0,0 ° Render EPSG:25832

The screenshot displays the QGIS desktop application. At the top, the title bar reads 'GIS-tutorial — QGIS'. Below it is a menu bar with options: Project, Edit, View, Layer, Settings, Plugins, and Vector. A toolbar with various icons for file operations, editing, and navigation is visible. The main window is divided into several panels. On the left, the 'Browser' panel is active, showing a tree view of data sources. It includes 'Favorites' (Spatial Bookmarks, Project Home, Home, C:\ (Windows-SSD), GeoPackage, SpatialLite) and 'XYZ Tiles' (Grabungsdokumentation.sqlite, PostgreSQL, SAP HANA, STAC, MS SQL Server, Oracle, WMS/WMTS, Cloud, Scenes, SensorThings, Vector Tiles, XYZ Tiles, Manzen Global Terrain). The 'Layers' panel at the bottom shows a single layer named 'OpenStreetMap' with a checkmark icon. The right side of the screen shows a map view of a green landscape with a river and a road.

GIS-tutoriel — QGIS

Project Edit View Layer Settings Plugins Vector

Browser

- ★ Favorites
- ▶ Spatial Bookmarks
- ▶ Project Home
- ▶ Home
- ▶ C:\ (Windows-SSD)
- ▶ GeoPackage
- ▶ Spatialite
- ▶ Grabungsdokumentation.sqlite
- ▶ PostgreSQL
- ▶ SAP HANA
- ▶ STAC
- ▶ MS SQL Server
- ▶ Oracle
- ▶ WMS/WMTS
- ▶ Cloud
- ▶ Scenes
- ▶ SensorThings
- ▶ Vector Tiles
- ▶ XYZ Tiles
- ▶ Manzen Global Terrain

Layers

- ✓ **OpenStreetMap**

Type to locate (Ctrl+K)

New Shapefile Layer

File name: C:\Users\jschm\Documents\Humanitees numerique\GIS-test\vector\villes.shp

File encoding: System

Geometry type: Point

Additional dimensions: ☒ None ☐ Z (+ M values) ☐ M values

Project CRS: EPSG:25832 - ETRS89 / UTM zone 32N

New Field

Name:

Type: 1.2 Decimal (double)

Length: 10 Precision: 6

Add to Fields List

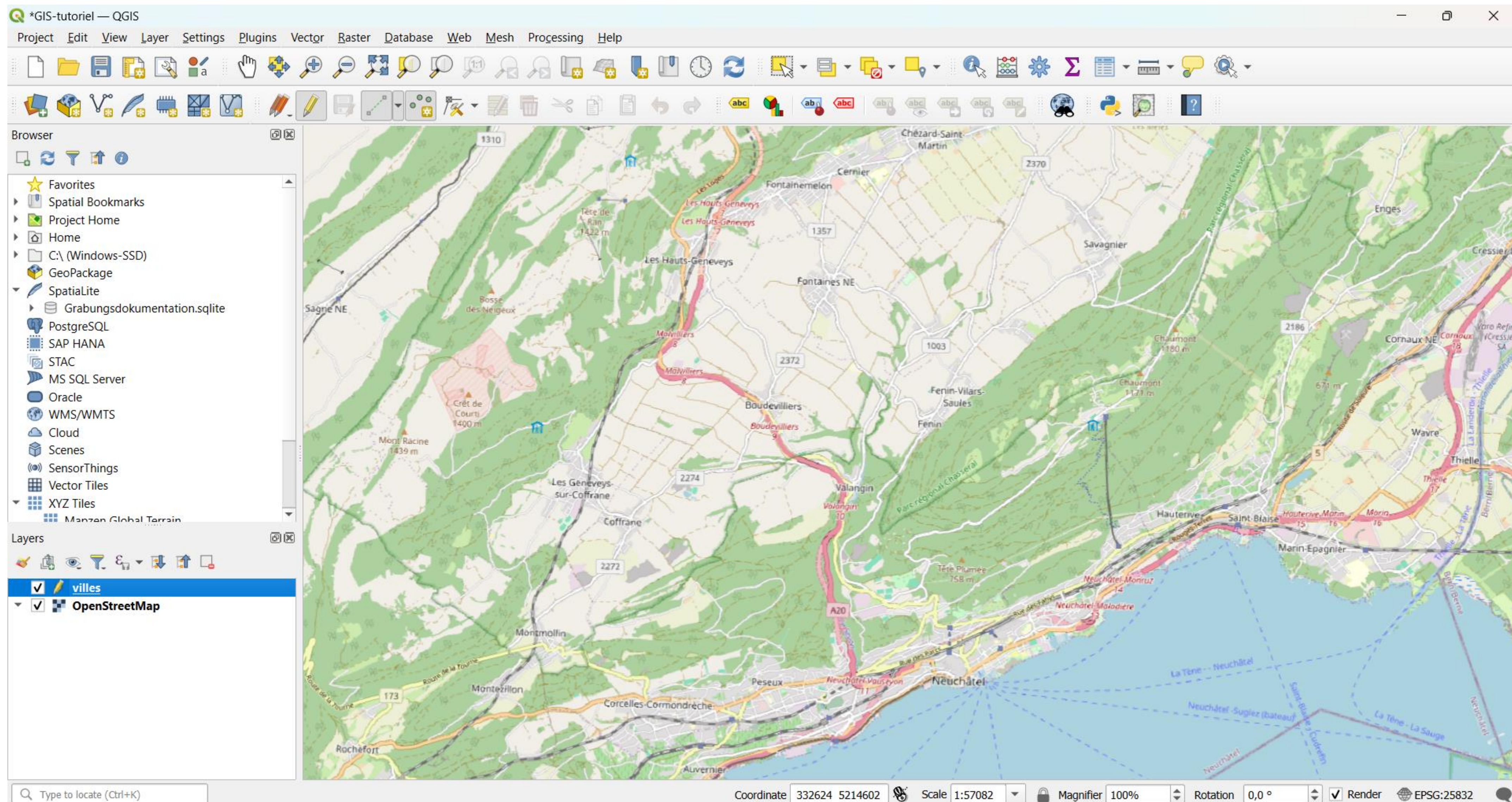
Fields List

Name	Type	Length	Precision
id	Integer	10	
Nom	String	80	
habitants	Integer	10	
elevation	Real	10	6

Remove Field

OK Cancel Help

Map view showing a topographic map of a region with labels like Enges, Cornaux NE, Wavre, Thielle, Saint-Blaise, Marin-Epagnier, and La Tène - La Saugie. The map is overlaid with a coordinate system (EPSG:25832) and a scale bar.



*GIS-tutoriel — QGIS

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

Browser

- ★ Favorites
- ▶ Spatial Bookmarks
- ▶ Project Home
- ▶ Home
- ▶ C:\ (Windows-SSD)
- ▶ GeoPackage
- ▶ SpatiaLite
- ▶ Grabungsdokumentation.sqlite
- ▶ PostgreSQL
- ▶ SAP HANA
- ▶ STAC
- ▶ MS SQL Server
- ▶ Oracle
- ▶ WMS/WMTS
- ▶ Cloud
- ▶ Scenes
- ▶ SensorThings
- ▶ Vector Tiles
- ▶ XYZ Tiles
- ▶ Manzen Global Terrain

Layers

- ✓ **villes**
- ✓ **OpenStreetMap**

ville...

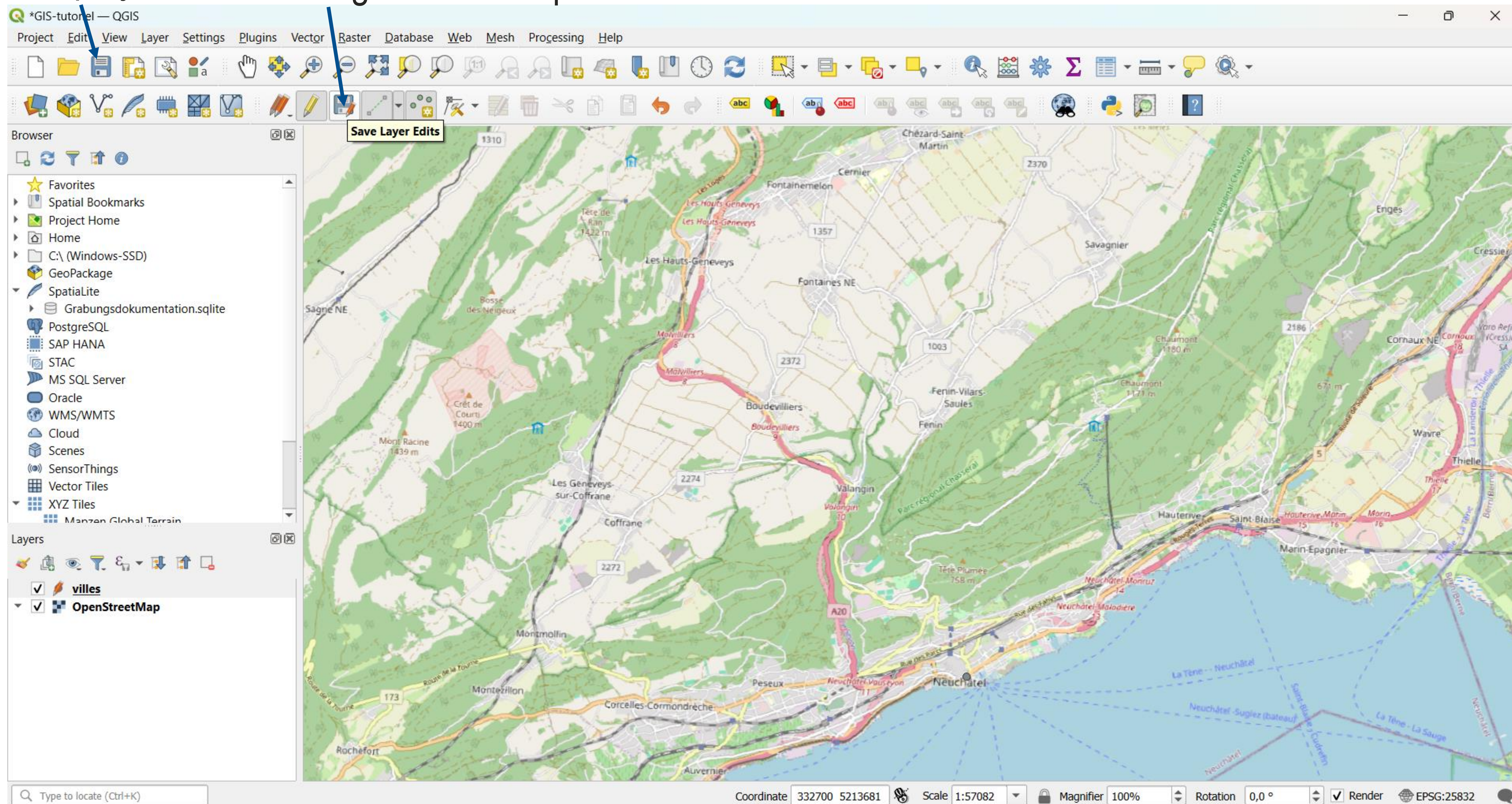
id	NULL
Nom	Neuchâtel
habitants	44898
elevation	434

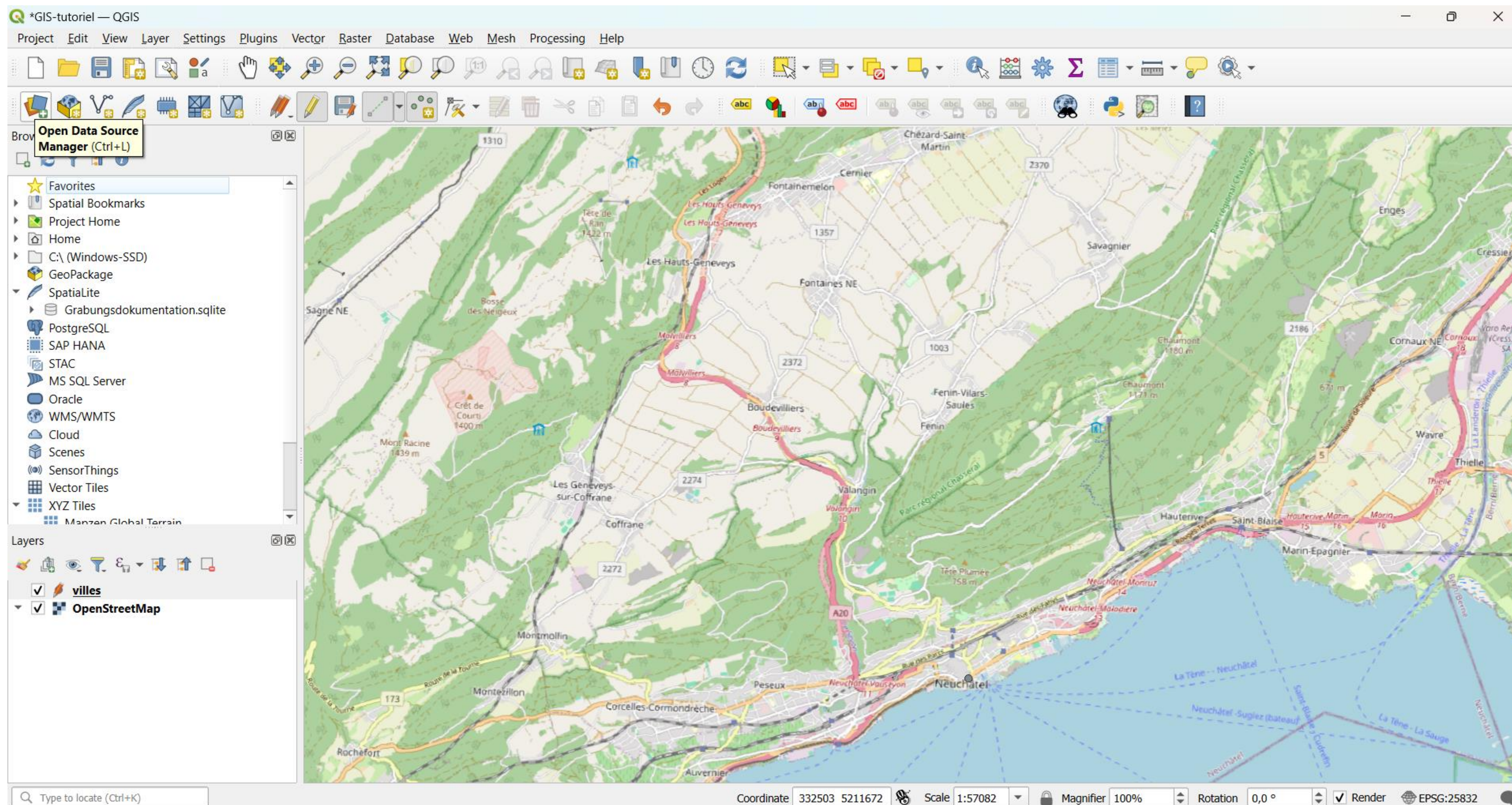
OK Cancel

Coordinate 342516 5204680 Scale 1:57082 Magnifier 100% Rotation 0,0 ° Render EPSG:25832

Sauvegarder
le projet

Sauvegarder le Shapefile





*GIS-tutorial — QGIS

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

Browser

- ★ Favorites
- ▶ Spatial Bookmarks
- ▶ Project Home
- ▶ Home
- ▶ C:\ (Windows-SSD)
- GeoPackage
- ▼ SpatialLite
 - Grabungsdokumentation.sqlite
- PostgreSQL
- SAP HANA
- STAC
- MS SQL Server
- Oracle
- WMS/WMTS
- Cloud
- Scenes
- SensorThings
- Vector Tiles
- ▼ XYZ Tiles
 - Manzan Global Terrain

Layers

- ✓ villes
- ▼ ✓ OpenStreetMap

Data Source Manager | XYZ

- GPS
- Spatialite
- PostgreSQL
- MS SQL Server
- Oracle
- Virtual Layer
- SAP HANA
- WMS/WMTS
- WFS / OGC API - Features
- WCS
- XYZ
- Vector Tile
- Scene
- ArcGIS REST Server
- SensorThings
- Metadata Search

XYZ Connections

Custom

New Edit Remove Load Save

Connection Details

URL

Authentication

Configurations Basic

Choose or create an authentication configuration

No Authentication

Configurations store encrypted credentials in the QGIS authentication database.

☐ Min. Zoom Level

☐ Max. Zoom Level

Referer

Tile Resolution

Interpretation

Close Add Help

Type to locate (Ctrl+K) 1 legend entry removed. Coordinate 392697 4970964 Scale 1:3152429 Magnifier 100% Rotation 0,0 ° Render EPSG:25832

*GIS-tutorial — QGIS

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

Browser

- ★ Favorites
- ▶ Spatial Bookmarks
- ▶ Project Home
- ▶ Home
- ▶ C:\ (Windows-SSD)
- ▶ GeoPackage
- ▶ SpatiaLite
- ▶ Grabungsdokumentation.sqlite
- ▶ PostgreSQL
- ▶ SAP HANA
- ▶ STAC
- ▶ MS SQL Server
- ▶ Oracle
- ▶ WMS/WMTS
- ▶ Cloud
- ▶ Scenes
- ▶ SensorThings
- ▶ Vector Tiles
- ▶ XYZ Tiles
- ▶ Manzen Global Terrain

Layers

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Data Source Manager | XYZ

XYZ Connections

XYZ Connection

Name: Google Earth

Connection Details

URL: `http://mt0.google.com/vt/lyrs=s&hl=en&x={x}&y={y}&z={z}`

Authentication

Configurations: Basic

Choose or create an authentication configuration

No Authentication

Configurations store encrypted credentials in the QGIS authentication database.

Min. Zoom Level: 0

Max. Zoom Level: 18

Referer:

Tile Resolution: Unknown (not scaled)

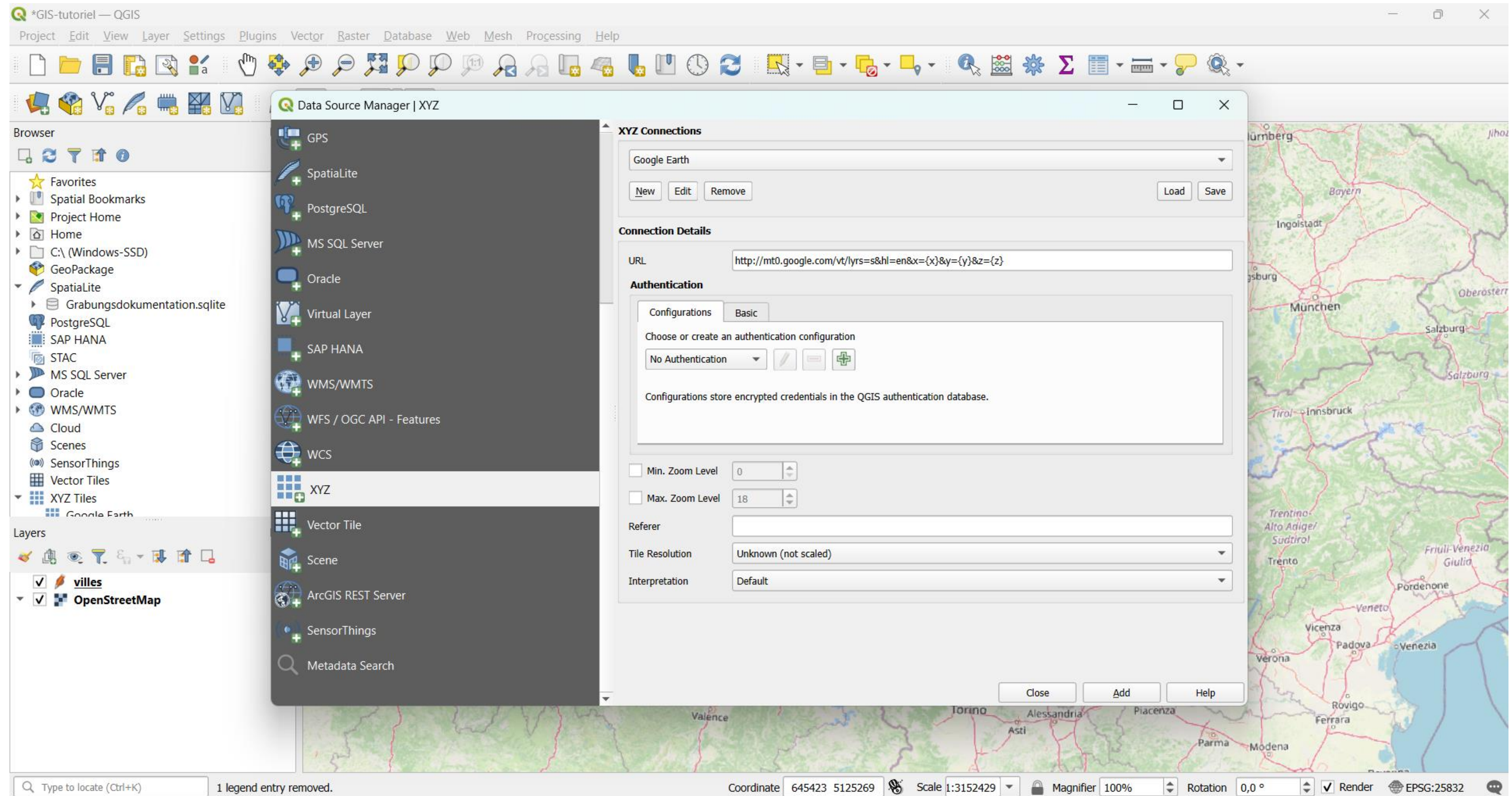
Interpretation: Default

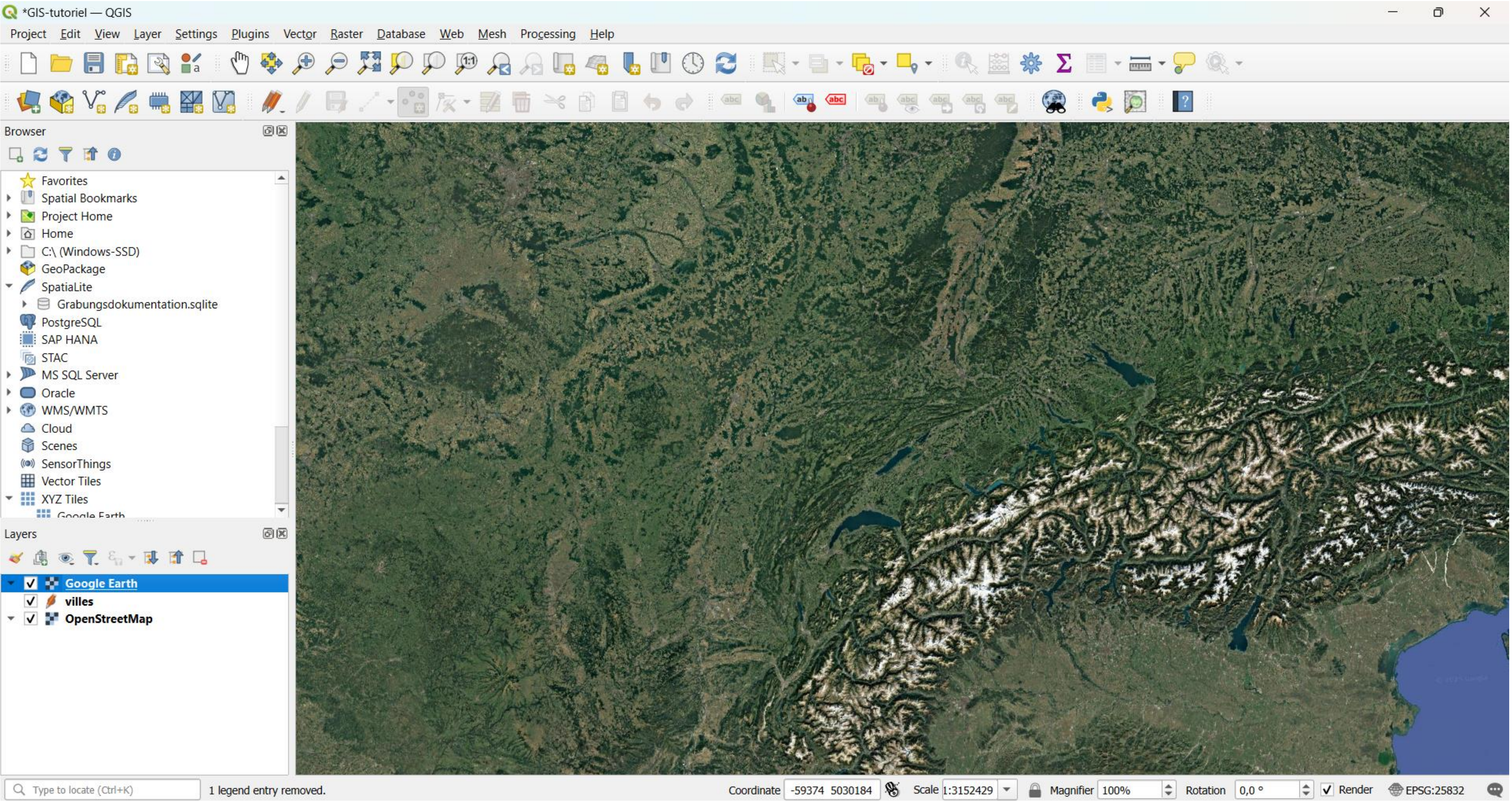
OK Cancel Help

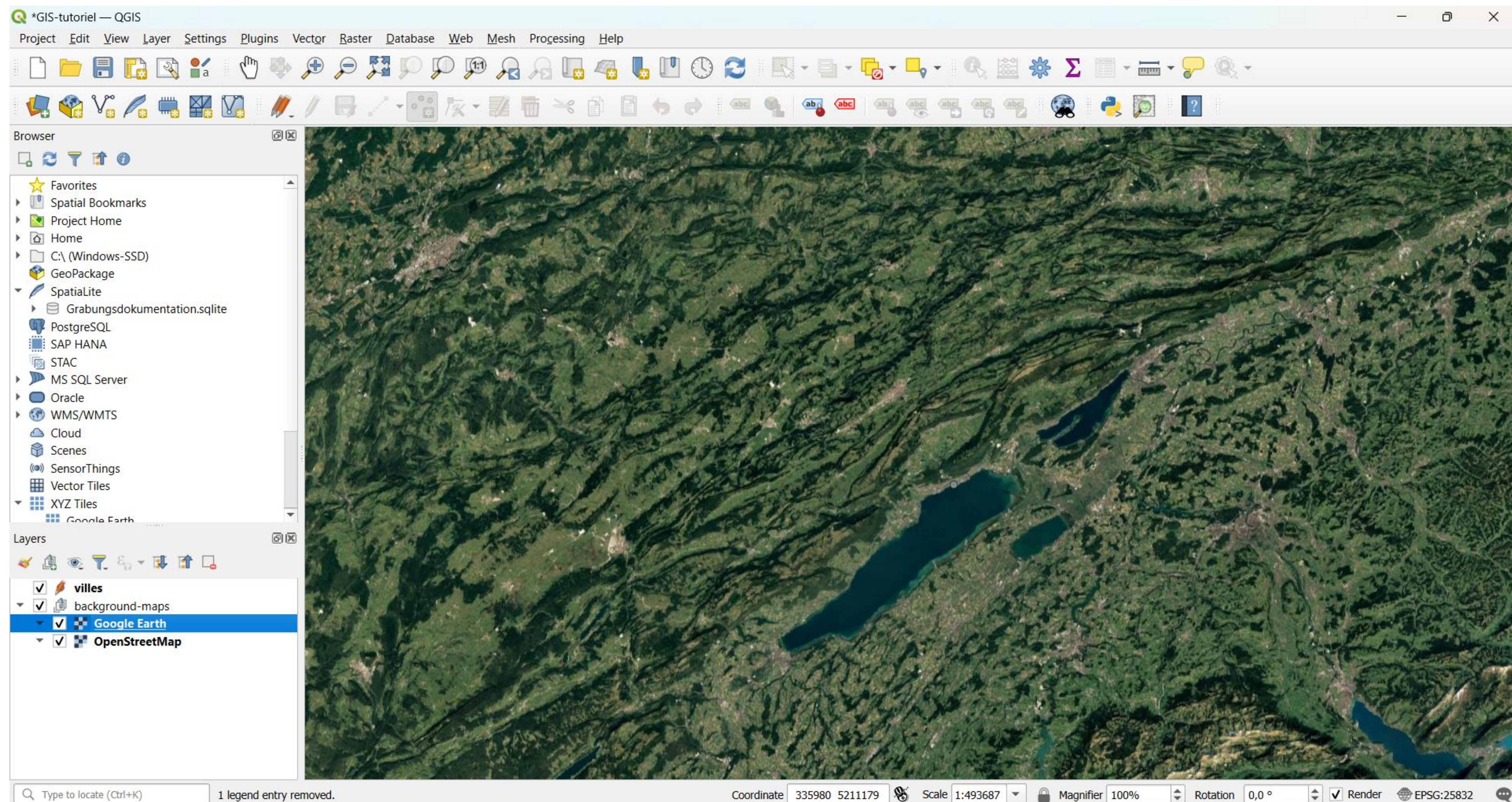
Close Add Help

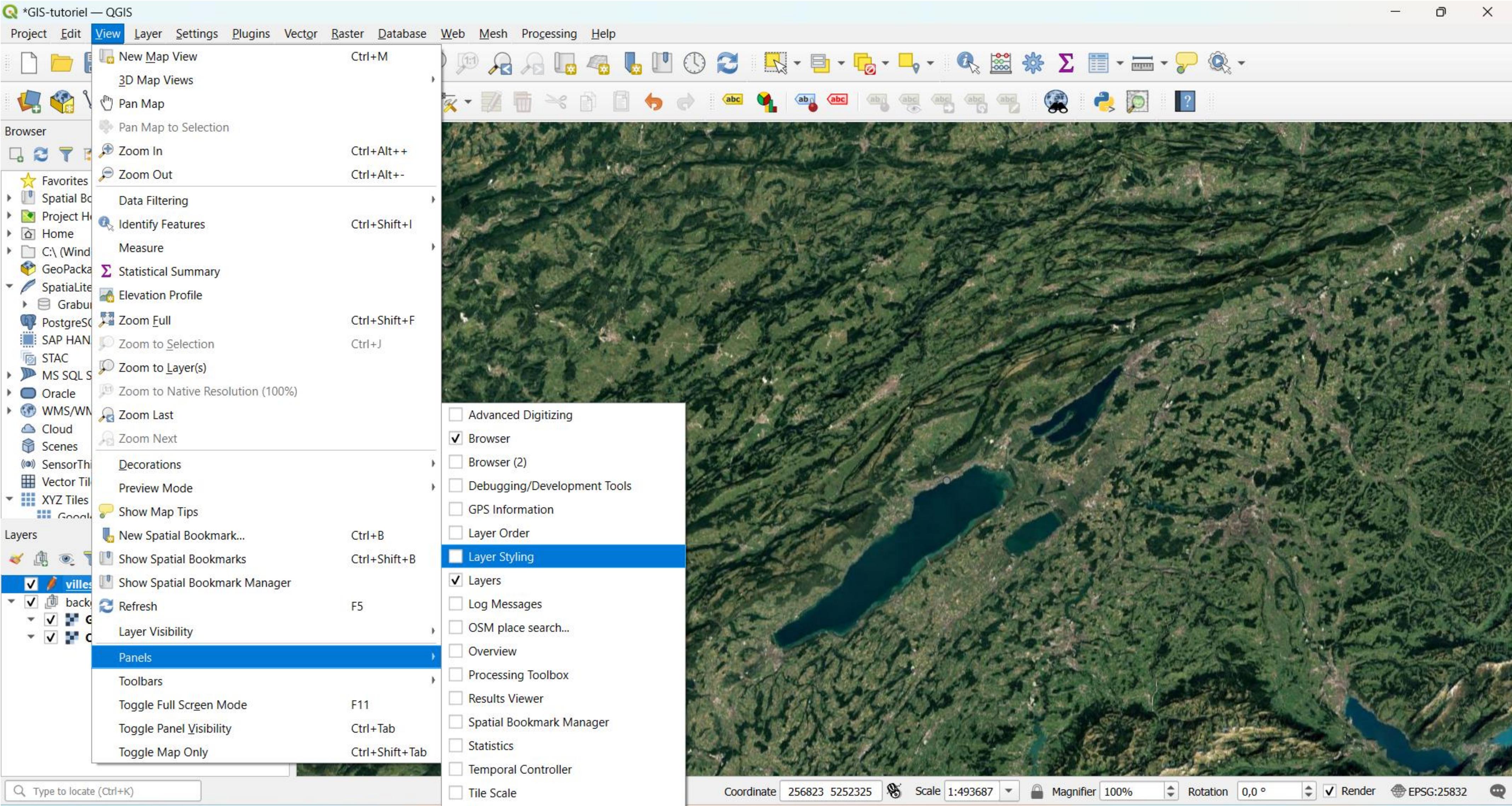
Type to locate (Ctrl+K) 1 legend entry removed. Coordinate: 645423 5125269 Scale: 1:3152429 Magnifier: 100% Rotation: 0,0 ° Render EPSG:25832

<https://hatarilabs.com/ih-en/how-to-add-a-google-map-in-qgis-3-tutorial>









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Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

Browser

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- ▶ WMS/WMTS
- ▶ Cloud
- ▶ Scenes
- ▶ SensorThings
- ▶ Vector Tiles
- ▶ XYZ Tiles
 - ▶ Google Earth

Layers

- ✓ **villes**
- ✓ background-maps
 - ✓ Google Earth
 - ✓ OpenStreetMap

Layer Styling

villes

Single Symbol

Marker

Simple Marker

Color

Opacity 100,0 %

Size 2,00000 Millimeters

Rotation 0,00 °

Favorites

Default

Project Styles

dot black

Save Symbol... Advanced

Layer Rendering

Live update Apply

Coordinate 385485 5228683 Scale 1:493687 Magnifier 100% Rotation 0,0 ° Render EPSG:25832

*GIS-tutorial — QGIS

Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help

Browser

- ★ Favorites
- ▶ Spatial Bookmarks
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- ▶ Vector Tiles
- ▶ XYZ Tiles
- ▶ Google Earth

Layers

- ✓ **villes**
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 - ✓ Google Earth
 - ✓ OpenStreetMap

Layer Styling

villes

Single Labels

Value: abc Nom

Font: Lato

Style: Regular

Size: 16,0000

Points

Color: [Black]

Opacity: 100,0 %

Allow HTML formatting

Search: Favorites

Default

Project Styles

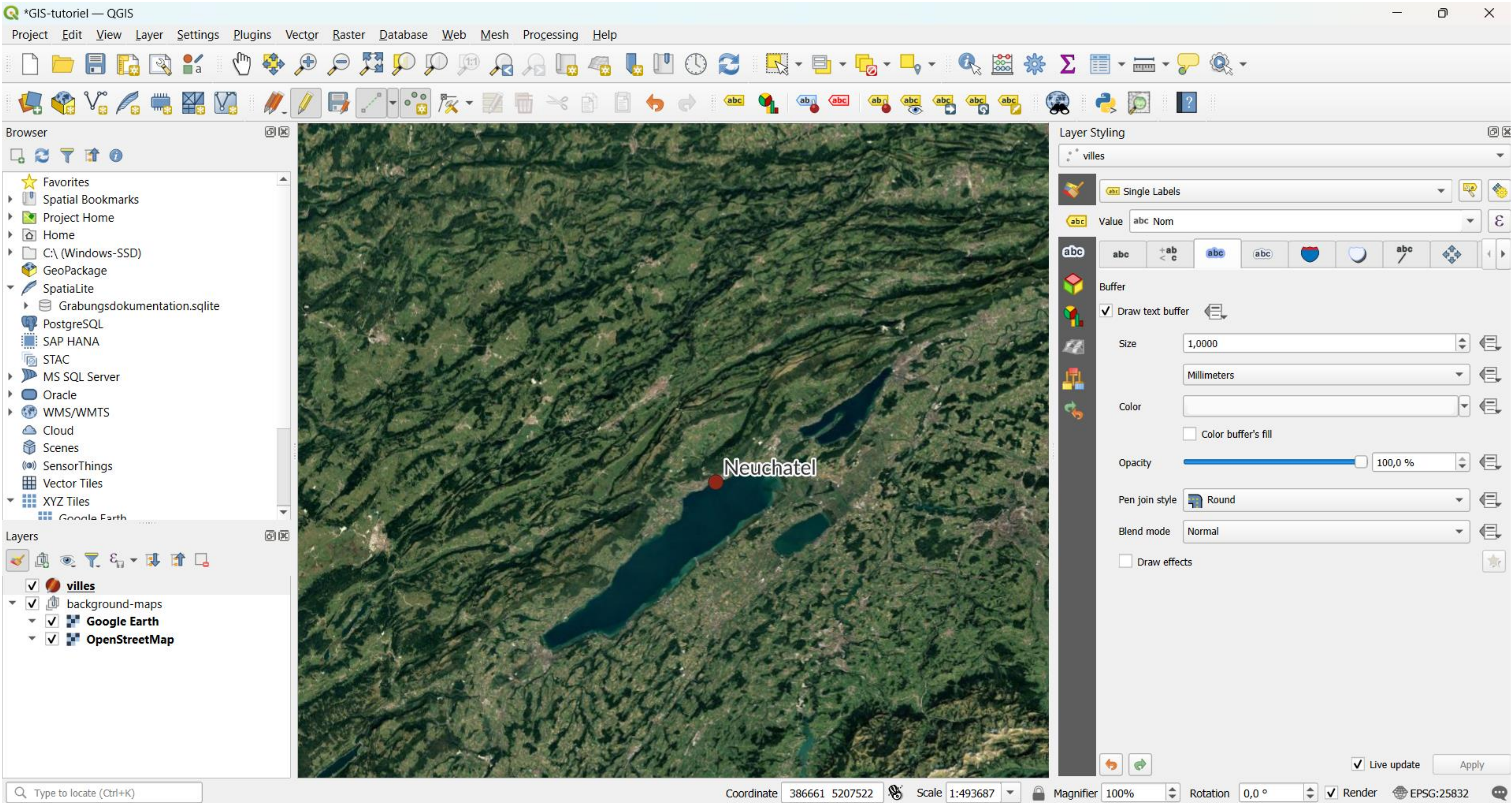
Aa

Save Settings...

Live update

Apply

Coordinate: 386661 5207522 Scale: 1:493687 Magnifier: 100% Rotation: 0,0 ° Render: EPSG:25832



Petit Pause – Questions?